

? Under what circumstances does the barbell push on the weightlifter just as hard as she pushes on the barbell? (i) When she holds the barbell stationary; (ii) when she raises the barbell; (iii) when she lowers the barbell; (iv) two of (i), (ii), and (iii); (v) all of (i), (ii), and (iii); (vi) none of these.



4 Newton's Laws of Motion

LEARNING OUTCOMES

In this chapter, you'll learn...

- 4.1 What the concept of force means in physics, why forces are vectors, and the significance of the net force on an object.
- 4.2 What happens when the net external force on an object is zero, and the significance of inertial frames of reference.
- 4.3 How the acceleration of an object is determined by the net external force on the object and the object's mass.
- 4.4 The difference between the mass of an object and its weight.
- 4.5 How the forces that two objects exert on each other are related.
- 4.6 How to use a free-body diagram to help analyze the forces on an object.

You'll need to review...

- 1.7 Vectors and vector addition.
- 1.8 Vector components.
- 2.4 Straight-line motion with constant acceleration.
- 2.5 The motion of freely falling objects.
- 3.2 Acceleration as a vector.
- 3.4 Uniform circular motion.
- 3.5 Relative velocity.

We've seen in the last two chapters how to use *kinematics* to describe motion in one, two, or three dimensions. But what *causes* objects to move the way that they do? For example, why does a dropped feather fall more slowly than a dropped baseball? Why do you feel pushed backward in a car that accelerates forward? The answers to such questions take us into the subject of **dynamics**, the relationship of motion to the forces that cause it.

The principles of dynamics were clearly stated for the first time by Sir Isaac Newton (1642–1727); today we call them **Newton's laws of motion**. Newton did not *derive* the laws of motion, but rather *deduced* them from a multitude of experiments performed by other scientists, especially Galileo Galilei (who died the year Newton was born). Newton's laws are the foundation of **classical mechanics** (also called **Newtonian mechanics**); using them, we can understand most familiar kinds of motion. Newton's laws need modification only for situations involving extremely high speeds (near the speed of light) or very small sizes (such as within the atom).

Newton's laws are very simple to state, yet many students find these laws difficult to grasp and to work with. The reason is that before studying physics, you've spent years walking, throwing balls, pushing boxes, and doing dozens of things that involve motion. Along the way, you've developed a set of “common sense” ideas about motion and its causes. But many of these “common sense” ideas don't stand up to logical analysis. A big part of the job of this chapter—and of the rest of our study of physics—is helping you recognize how “common sense” ideas can sometimes lead you astray, and how to adjust your understanding of the physical world to make it consistent with what experiments tell us.

4.1 FORCE AND INTERACTIONS

A **force** is a push or a pull. More precisely, a force is an *interaction* between two objects or between an object and its environment (**Fig. 4.1**). That's why we always refer to the force that one object *exerts* on a second object. When you push on a car that is stuck in the snow, you exert a force on the car; a steel cable exerts a force on the beam it is hoisting at a construction site; and so on. As Fig. 4.1 shows, force is a *vector* quantity; you can push or pull an object in different directions.

When a force involves direct contact between two objects, such as a push or pull that you exert on an object with your hand, we call it a **contact force**. **Figures 4.2a, 4.2b, and 4.2c** show three common types of contact forces. The **normal force** (Fig. 4.2a) is exerted on an object by any surface with which it is in contact. The adjective “normal” means that the force always acts *perpendicular* to the surface of contact, no matter what the angle of that surface. By contrast, the **friction force** (Fig. 4.2b) exerted on an object by a surface acts *parallel* to the surface, in the direction that opposes sliding. The pulling force exerted by a stretched rope or cord on an object to which it’s attached is called a **tension force** (Fig. 4.2c). When you tug on your dog’s leash, the force that pulls on her collar is a tension force.

In addition to contact forces, there are **long-range forces** that act even when the objects are separated by empty space. The force between two magnets is an example of a long-range force, as is the force of gravity (Fig. 4.2d); the earth pulls a dropped object toward it even though there is no direct contact between the object and the earth. The gravitational force that the earth exerts on your body is called your **weight**.

To describe a force vector $\vec{F}$, we need to describe the *direction* in which it acts as well as its *magnitude*, the quantity that describes “how much” or “how hard” the force pushes or pulls. The SI unit of the magnitude of force is the *newton*, abbreviated N. (We’ll give a precise definition of the newton in Section 4.3.) **Table 4.1** lists some typical force magnitudes.

A common instrument for measuring force magnitudes is the *spring balance*. It consists of a coil spring enclosed in a case with a pointer attached to one end. When forces are applied to the ends of the spring, it stretches by an amount that depends on the force. We can make a scale for the pointer by using a number of identical objects with weights of exactly 1 N each. When one, two, or more of these are suspended simultaneously from the balance, the total force stretching the spring is 1 N, 2 N, and so on, and we can label the corresponding positions of the pointer 1 N, 2 N, and so on. Then we can use this instrument to measure the magnitude of an unknown force. We can also make a similar instrument that measures pushes instead of pulls.

Figure 4.3 (next page) shows a spring balance being used to measure a pull or push that we apply to a box. In each case we draw a vector to represent the applied force. The length of the vector shows the magnitude; the longer the vector, the greater the force magnitude.

Superposition of Forces

When you hold a ball in your hand to throw it, at least two forces act on it: the push of your hand and the downward pull of gravity. Experiment shows that when two forces $\vec{F}_1$ and $\vec{F}_2$ act at the same time at the same point on an object (**Fig. 4.4**, next page), the effect on the object’s motion is the same as if a single force $\vec{R}$ were acting equal to the vector sum, or resultant, of the original forces: $\vec{R} = \vec{F}_1 + \vec{F}_2$. More generally, *any number of*

Figure 4.1 Some properties of forces.

- A force is a push or a pull.
- A force is an interaction between two objects or between an object and its environment.
- A force is a vector quantity, with magnitude and direction.

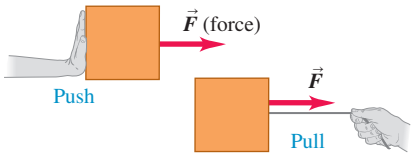
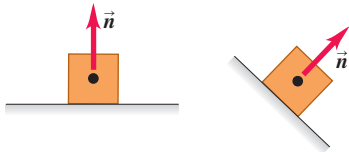
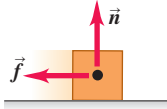


Figure 4.2 Four common types of forces.

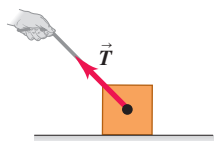
(a) **Normal force $\vec{n}$** : When an object rests or pushes on a surface, the surface exerts a push on it that is directed perpendicular to the surface.



(b) **Friction force $\vec{f}$** : In addition to the normal force, a surface may exert a friction force on an object, directed parallel to the surface.



(c) **Tension force $\vec{T}$** : A pulling force exerted on an object by a rope, cord, etc.



(d) **Weight $\vec{w}$** : The pull of gravity on an object is a long-range force (a force that acts over a distance).

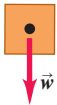
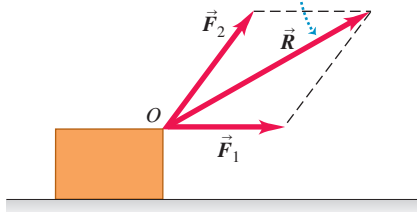


TABLE 4.1 Typical Force Magnitudes

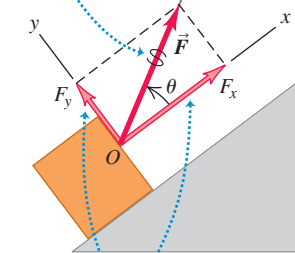
Sun’s gravitational force on the earth	3.5×10^{22} N
Weight of a large blue whale	1.9×10^6 N
Maximum pulling force of a locomotive	8.9×10^5 N
Weight of a 250 lb linebacker	1.1×10^3 N
Weight of a medium apple	1 N
Weight of the smallest insect eggs	2×10^{-6} N
Electric attraction between the proton and the electron in a hydrogen atom	8.2×10^{-8} N
Weight of a very small bacterium	1×10^{-18} N
Weight of a hydrogen atom	1.6×10^{-26} N
Weight of an electron	8.9×10^{-30} N
Gravitational attraction between the proton and the electron in a hydrogen atom	3.6×10^{-47} N

Figure 4.4 Superposition of forces.

Two forces $\vec{F}_1$ and $\vec{F}_2$ acting on an object at point O have the same effect as a single force $\vec{R}$ equal to their vector sum.

Figure 4.5 F_x and F_y are the components of $\vec{F}$ parallel and perpendicular to the sloping surface of the inclined plane.

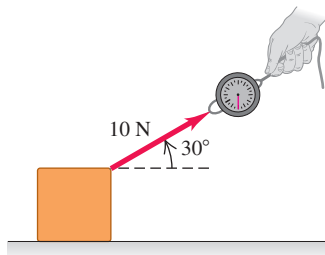
We cross out a vector when we replace it by its components.



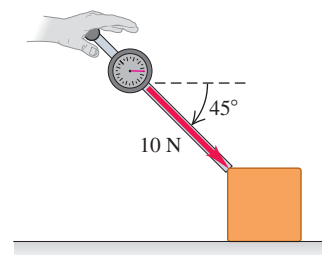
The x - and y -axes can have any orientation, just so they're mutually perpendicular.

Figure 4.3 Using a vector arrow to denote the force that we exert when (a) pulling a block with a string or (b) pushing a block with a stick.

(a) A 10 N pull directed 30° above the horizontal



(b) A 10 N push directed 45° below the horizontal



forces applied at a point on an object have the same effect as a single force equal to the vector sum of the forces. This important principle is called **superposition of forces**.

Since forces are vector quantities and add like vectors, we can use all of the rules of vector mathematics that we learned in Chapter 1 to solve problems that involve vectors. This would be a good time to review the rules for vector addition presented in Sections 1.7 and 1.8.

We learned in Section 1.8 that it's easiest to add vectors by using components. That's why we often describe a force $\vec{F}$ in terms of its x - and y -components F_x and F_y . Note that the x - and y -coordinate axes do *not* have to be horizontal and vertical, respectively. As an example, **Fig. 4.5** shows a crate being pulled up a ramp by a force $\vec{F}$. In this situation it's most convenient to choose one axis to be parallel to the ramp and the other to be perpendicular to the ramp. For the case shown in Fig. 4.5, both F_x and F_y are positive; in other situations, depending on your choice of axes and the orientation of the force $\vec{F}$, either F_x or F_y may be negative or zero.

CAUTION Using a wiggly line in force diagrams In Fig. 4.5 we draw a wiggly line through the force vector $\vec{F}$ to show that we have replaced it by its x - and y -components. Otherwise, the diagram would include the same force twice. We'll draw such a wiggly line in any force diagram where a force is replaced by its components. We encourage you to do the same in your own diagrams! ■

We'll often need to find the vector sum (resultant) of *all* forces acting on an object. We call this the **net force** acting on the object. We'll use the Greek letter Σ (capital sigma, equivalent to the Roman S) as a shorthand notation for a sum. If the forces are labeled $\vec{F}_1$, $\vec{F}_2$, $\vec{F}_3$, and so on, we can write

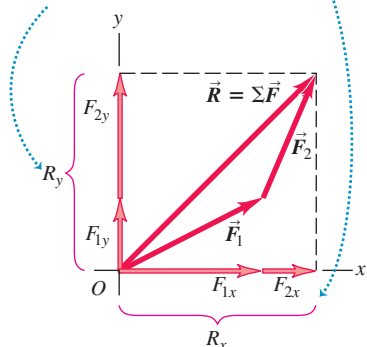
$$\text{The net force acting on an object } \dots \vec{R} = \Sigma \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots \quad (4.1)$$

... is the vector sum, or resultant, of all individual forces acting on that object.

Figure 4.6 Finding the components of the vector sum (resultant) $\vec{R}$ of two forces $\vec{F}_1$ and $\vec{F}_2$.

The y -component of $\vec{R}$ equals the sum of the y -components of $\vec{F}_1$ and $\vec{F}_2$.

The same is true for the x -components.



We read $\Sigma \vec{F}$ as “the vector sum of the forces” or “the net force.” The x -component of the net force is the sum of the x -components of the individual forces, and likewise for the y -component (**Fig. 4.6**):

$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad (4.2)$$

Each component may be positive or negative, so be careful with signs when you evaluate these sums.

Once we have R_x and R_y we can find the magnitude and direction of the net force $\vec{R} = \Sigma \vec{F}$ acting on the object. The magnitude is

$$R = \sqrt{R_x^2 + R_y^2}$$

and the angle θ between $\vec{R}$ and the $+x$ -axis can be found from the relationship $\tan \theta = R_y/R_x$. The components R_x and R_y may be positive, negative, or zero, and the angle θ may be in any of the four quadrants.

In three-dimensional problems, forces may also have z -components; then we add the equation $R_z = \sum F_z$ to Eqs. (4.2). The magnitude of the net force is then

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2}$$

EXAMPLE 4.1 Superposition of forces: Finding the net force

WITH **✓** VARIATION PROBLEMS

Three professional wrestlers are fighting over a champion's belt. **Figure 4.7a** shows the horizontal force each wrestler applies to the belt, as viewed from above. The forces have magnitudes $F_1 = 50$ N, $F_2 = 120$ N, and $F_3 = 250$ N. Find the x - and y -components of the net force on the belt, and find its magnitude and direction.

IDENTIFY and SET UP This is a problem in vector addition in which the vectors happen to represent forces. To find the x - and y -components of the net force $\vec{R}$, we'll use the component method of vector addition expressed by Eqs. (4.2). Once we know the components of $\vec{R}$, we can find its magnitude and direction.

EXECUTE Figure 4.7a shows that force $\vec{F}_1$ (magnitude 50 N) points in the positive x -direction. Hence it has a positive x -component and zero y -component:

$$F_{1x} = 50 \text{ N} \quad F_{1y} = 0 \text{ N}$$

Force $\vec{F}_2$ points in the negative y -direction and so has zero x -component and a negative y -component:

$$F_{2x} = 0 \text{ N} \quad F_{2y} = -120 \text{ N}$$

Force $\vec{F}_3$ doesn't point along either the x -direction or the y -direction: Figure 4.7a shows that its x -component is negative and its y -component

is positive. The angle between $\vec{F}_3$ and the negative x -axis is 53° , so the absolute value of its x -component is equal to the magnitude of $\vec{F}_3$ times the cosine of 53° . The absolute value of the y -component is therefore the magnitude of $\vec{F}_3$ times the sine of 53° . Keeping track of the signs, we find the components of $\vec{F}_3$ are

$$F_{3x} = -(250 \text{ N}) \cos 53^\circ = -150 \text{ N} \quad F_{3y} = (250 \text{ N}) \sin 53^\circ = 200 \text{ N}$$

From Eqs. (4.2) the components of the net force $\vec{R} = \sum \vec{F}$ are

$$R_x = F_{1x} + F_{2x} + F_{3x} = 50 \text{ N} + 0 \text{ N} + (-150 \text{ N}) = -100 \text{ N}$$

$$R_y = F_{1y} + F_{2y} + F_{3y} = 0 \text{ N} + (-120 \text{ N}) + 200 \text{ N} = 80 \text{ N}$$

The net force has a negative x -component and a positive y -component, as Fig. 4.7b shows.

The magnitude of $\vec{R}$ is

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(-100 \text{ N})^2 + (80 \text{ N})^2} = 128 \text{ N}$$

To find the angle between the net force and the $+x$ -axis, we use Eq. (1.7):

$$\theta = \arctan \frac{R_y}{R_x} = \arctan \left(\frac{80 \text{ N}}{-100 \text{ N}} \right) = \arctan(-0.80)$$

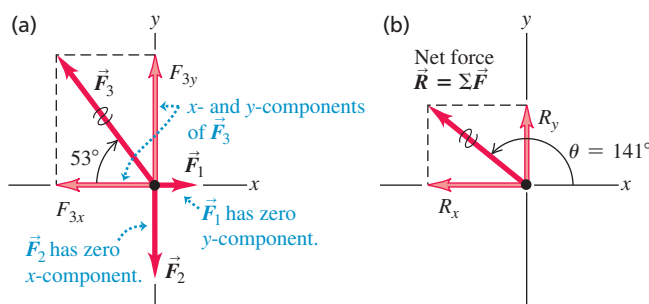
The arctangent of -0.80 is -39° , but Fig. 4.7b shows that the net force lies in the second quadrant. Hence the correct solution is $\theta = -39^\circ + 180^\circ = 141^\circ$.

EVALUATE The net force is *not* zero. Wrestler 3 exerts the greatest force on the belt, $F_3 = 250$ N, and will walk away with it when the struggle ends.

You should check the direction of $\vec{R}$ by adding the vectors $\vec{F}_1$, $\vec{F}_2$, and $\vec{F}_3$ graphically. Does your drawing show that $\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$ points in the second quadrant as we found?

KEYCONCEPT The net force is the vector sum of all of the individual forces that act on the object. It can be specified by its components or by its magnitude and direction.

Figure 4.7 (a) Three forces acting on a belt. (b) The net force $\vec{R} = \sum \vec{F}$ and its components.



TEST YOUR UNDERSTANDING OF SECTION 4.1 Figure 4.5 shows a force $\vec{F}$ acting on a crate. (a) With the x - and y -axes shown in the figure, is the x -component of the gravitational force that the earth exerts on the crate (the crate's weight) positive, negative, or zero? (b) What about the y -component?

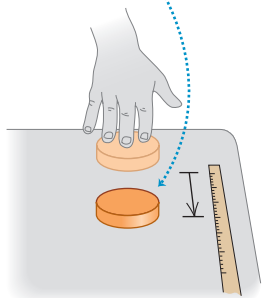
ANSWER (a) **negative.** The gravitational force on the crate points straight downward. In Fig. 4.5 the x -axis points up and to the right, and the y -axis points up and to the left. With this choice of axes, the gravitational force has both an x -component and a y -component, and both are negative.

4.2 NEWTON'S FIRST LAW

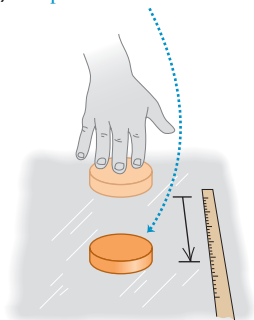
How do the forces acting on an object affect that object's motion? Let's first note that it's impossible for an object to affect its own motion by exerting a force on itself. If that were possible, you could lift yourself to the ceiling by pulling up on your belt! The forces that affect an object's motion are **external forces**, those forces exerted on the object by other

Figure 4.8 The slicker the surface, the farther a puck slides after being given an initial velocity. On an air-hockey table (c) the friction force is practically zero, so the puck continues with almost constant velocity.

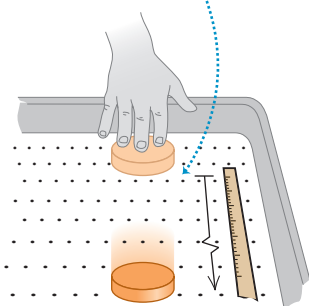
(a) Table: puck stops short.



(b) Ice: puck slides farther.



(c) Air-hockey table: puck slides even farther.



objects in its environment. So the question we must answer is this: How do the external forces that act on an object affect its motion?

To begin to answer this question, let's first consider what happens when the net external force on an object is *zero*. You would almost certainly agree that if an object is at rest, and if no net external force acts on it (that is, no net push or pull from other objects), that object will remain at rest. But what if there is zero net external force acting on an object in *motion*?

To see what happens in this case, suppose you slide a hockey puck along a horizontal tabletop, applying a horizontal force to it with your hand (Fig. 4.8a). After you stop pushing, the puck does *not* continue to move indefinitely; it slows down and stops. To keep it moving, you have to keep pushing (that is, applying a force). You might come to the “common sense” conclusion that objects in motion naturally come to rest and that a force is required to sustain motion.

But now imagine pushing the puck across a smooth surface of ice (Fig. 4.8b). After you quit pushing, the puck will slide a lot farther before it stops. Put it on an air-hockey table, where it floats on a thin cushion of air, and it moves still farther (Fig. 4.8c). In each case, what slows the puck down is *friction*, an interaction between the lower surface of the puck and the surface on which it slides. Each surface exerts a friction force on the puck that resists the puck's motion; the difference in the three cases is the magnitude of the friction force. The ice exerts less friction than the tabletop, so the puck travels farther. The gas molecules of the air-hockey table exert the least friction of all. If we could eliminate friction completely, the puck would never slow down, and we would need no force at all to keep the puck moving once it had been started. Thus the “common sense” idea that a force is required to sustain motion is *incorrect*.

Experiments like the ones we've just described show that when *no* net external force acts on an object, the object either remains at rest *or* moves with constant velocity in a straight line. Once an object has been set in motion, no net external force is needed to keep it moving. We call this observation *Newton's first law of motion*:

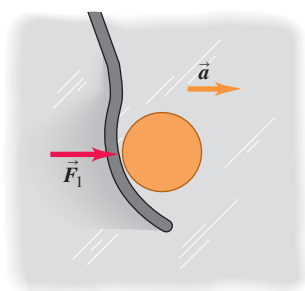
NEWTON'S FIRST LAW OF MOTION An object acted on by no net external force has a constant velocity (which may be zero) and zero acceleration.

The tendency of an object to keep moving once it is set in motion is called **inertia**. You use inertia when you try to get ketchup out of a bottle by shaking it. First you start the bottle (and the ketchup inside) moving forward; when you jerk the bottle back, the ketchup tends to keep moving forward and, you hope, ends up on your burger. Inertia is also the tendency of an object at rest to remain at rest. You may have seen a tablecloth yanked out from under a table setting without breaking anything. The force on the table setting isn't great enough to make it move appreciably during the short time it takes to pull the tablecloth away.

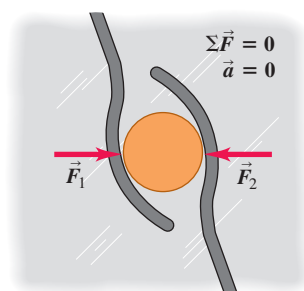
It's important to note that the *net* external force is what matters in Newton's first law. For example, a physics book at rest on a horizontal tabletop has two forces acting on it: an upward supporting force, or normal force, exerted by the tabletop (see Fig. 4.2a) and the downward force of the earth's gravity (see Fig. 4.2d). The upward push of the surface is just as great as the downward pull of gravity, so the *net* external force acting on the book (that is, the vector sum of the two forces) is zero. In agreement with Newton's first law, if the book is at rest on the tabletop, it remains at rest. The same principle applies to a hockey puck sliding on a horizontal, frictionless surface: The vector sum of the upward push of the surface and the downward pull of gravity is zero. Once the puck is in motion, it continues to move with constant velocity because the *net* external force acting on it is zero.

Here's another example. Suppose a hockey puck rests on a horizontal surface with negligible friction, such as an air-hockey table or a slab of wet ice. If the puck is initially at rest and a single horizontal force $\vec{F}_1$ acts on it (Fig. 4.9a), the puck starts to move. If the puck is in motion to begin with, the force changes its speed, its direction, or both, depending on the direction of the force. In this case the net external force is equal to $\vec{F}_1$, which is *not* zero. (There are also two vertical forces: the earth's gravitational attraction and the upward normal force exerted by the surface. But as we mentioned earlier, these two forces cancel.)

(a) A puck on a frictionless surface accelerates when acted on by a single horizontal force.



(b) This puck is acted on by two horizontal forces whose vector sum is zero. The puck behaves as though no external forces act on it.



Now suppose we apply a second force, $\vec{F}_2$ (Fig. 4.9b), equal in magnitude to $\vec{F}_1$ but opposite in direction. The two forces are negatives of each other, $\vec{F}_2 = -\vec{F}_1$, and their vector sum is zero:

$$\Sigma \vec{F} = \vec{F}_1 + \vec{F}_2 = \vec{F}_1 + (-\vec{F}_1) = 0$$

Again, we find that if the object is at rest at the start, it remains at rest; if it is initially moving, it continues to move in the same direction with constant speed. These results show that in Newton's first law, *zero net external force is equivalent to no external force at all*. This is just the principle of superposition of forces that we saw in Section 4.1.

When an object is either at rest or moving with constant velocity (in a straight line with constant speed), we say that the object is in **equilibrium**. For an object to be in equilibrium, it must be acted on by no forces, or by several forces such that their vector sum—that is, the net external force—is zero:

Newton's first law:
Net external force on an object ... $\rightarrow \Sigma \vec{F} = 0 \leftarrow$... must be zero if the object is in equilibrium. (4.3)

We're assuming that the object can be represented adequately as a point particle. When the object has finite size, we also have to consider *where* on the object the forces are applied. We'll return to this point in Chapter 11.

CONCEPTUAL EXAMPLE 4.2 Using Newton's first law I

In the classic 1950 science-fiction film *Rocketship X-M*, a spaceship is moving in the vacuum of outer space, far from any star or planet, when its engine dies. As a result, the spaceship slows down and stops. What does Newton's first law say about this scene?

SOLUTION No external forces act on the spaceship after the engine dies, so according to Newton's first law it will *not* stop but will continue

to move in a straight line with constant speed. Some science-fiction movies are based on accurate science; this is not one of them.

KEYCONCEPT If the net external force on an object is zero, the object either remains at rest or keeps moving at a constant velocity.

CONCEPTUAL EXAMPLE 4.3 Using Newton's first law II

You are driving a Porsche 918 Spyder on a straight testing track at a constant speed of 250 km/h. You pass a 1971 Volkswagen Beetle doing a constant 75 km/h. On which car is the net external force greater?

SOLUTION The key word in this question is “net.” Both cars are in equilibrium because their velocities are constant; Newton's first law therefore says that the *net* external force on each car is *zero*.

This seems to contradict the “common sense” idea that the faster car must have a greater force pushing it. Thanks to your Porsche's high-power engine, it's true that the track exerts a greater forward force on

your Porsche than it does on the Volkswagen. But a *backward* force also acts on each car due to road friction and air resistance. When the car is traveling with constant velocity, the vector sum of the forward and backward forces is zero. There is a greater backward force on the fast-moving Porsche than on the slow-moving Volkswagen, which is why the Porsche's engine must be more powerful than that of the Volkswagen.

KEYCONCEPT If an object either remains at rest or keeps moving at a constant velocity, the net external force on the object is zero.

Figure 4.9 (a) A hockey puck accelerates in the direction of a net applied force $\vec{F}_1$. (b) When the net external force is zero, the acceleration is zero, and the puck is in equilibrium.

APPLICATION Sledding with

Newton's First Law The downward force of gravity acting on the child and sled is balanced by an upward normal force exerted by the ground. The adult's foot exerts a forward force that balances the backward force of friction on the sled. Hence there is no net external force on the child and sled, and they slide with a constant velocity.



Inertial Frames of Reference

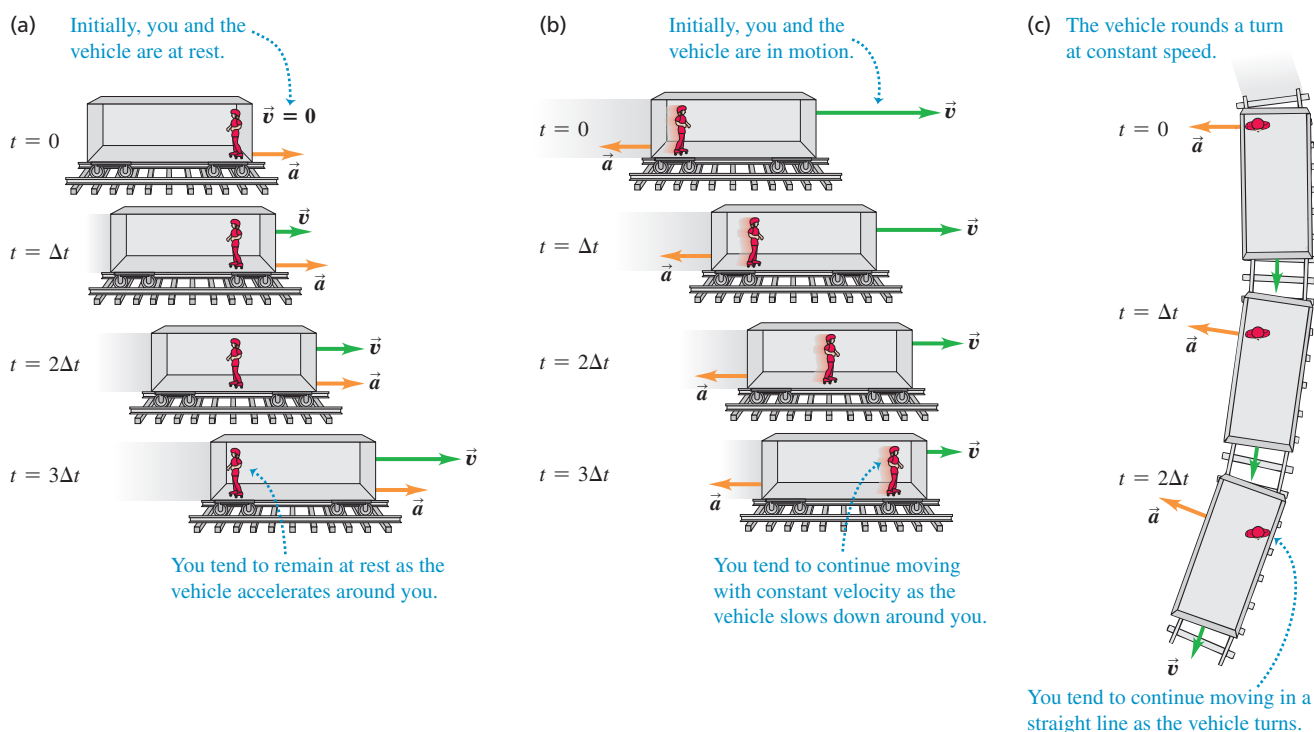
In discussing relative velocity in Section 3.5, we introduced the concept of *frame of reference*. This concept is central to Newton's laws of motion. Suppose you are in a bus that is traveling on a straight road and speeding up. If you could stand in the aisle on roller skates, you would start moving *backward* relative to the bus as the bus gains speed. If instead the bus was slowing to a stop, you would start moving *forward* down the aisle. In either case, it looks as though Newton's first law is not obeyed; there is no net external force acting on you, yet your velocity changes. What's wrong?

The point is that the bus is accelerating with respect to the earth and is *not* a suitable frame of reference for Newton's first law. This law is valid in some frames of reference and not valid in others. A frame of reference in which Newton's first law *is* valid is called an **inertial frame of reference**. The earth is at least approximately an inertial frame of reference, but the bus is not. (The earth is not a completely inertial frame, owing to the acceleration associated with its rotation and its motion around the sun. These effects are quite small, however; see Exercises 3.25 and 3.34.) Because Newton's first law is used to define what we mean by an inertial frame of reference, it is sometimes called the *law of inertia*.

Figure 4.10 helps us understand what you experience when riding in a vehicle that's accelerating. In Fig. 4.10a, a vehicle is initially at rest and then begins to accelerate to the right. A passenger standing on roller skates (which nearly eliminate the effects of friction) has virtually no net external force acting on her, so she tends to remain at rest relative to the inertial frame of the earth. As the vehicle accelerates around her, she moves backward relative to the vehicle. In the same way, a passenger in a vehicle that is slowing down tends to continue moving with constant velocity relative to the earth, and so moves forward relative to the vehicle (Fig. 4.10b). A vehicle is also accelerating if it moves at a constant speed but is turning (Fig. 4.10c). In this case a passenger tends to continue moving relative to the earth at constant speed in a straight line; relative to the vehicle, the passenger moves to the side of the vehicle on the outside of the turn.

In each case shown in Fig. 4.10, an observer in the vehicle's frame of reference might be tempted to conclude that there *is* a net external force acting on the passenger, since the passenger's velocity *relative to the vehicle* changes in each case. This conclusion is simply

Figure 4.10 Riding in an accelerating vehicle.



wrong; the net external force on the passenger is indeed zero. The vehicle observer's mistake is in trying to apply Newton's first law in the vehicle's frame of reference, which is *not* an inertial frame and in which Newton's first law isn't valid (**Fig. 4.11**). In this book we'll use *only* inertial frames of reference.

We've mentioned only one (approximately) inertial frame of reference: the earth's surface. But there are many inertial frames. If we have an inertial frame of reference A , in which Newton's first law is obeyed, then *any* second frame of reference B will also be inertial if it moves relative to A with constant velocity $\vec{v}_{B/A}$. We can prove this by using the relative-velocity relationship Eq. (3.35) from Section 3.5:

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$$

Suppose that P is an object that moves with constant velocity $\vec{v}_{P/A}$ with respect to an inertial frame A . By Newton's first law the net external force on this object is zero. The velocity of P relative to another frame B has a different value, $\vec{v}_{P/B} = \vec{v}_{P/A} - \vec{v}_{B/A}$. But if the relative velocity $\vec{v}_{B/A}$ of the two frames is constant, then $\vec{v}_{P/B}$ is constant as well. Thus B is also an inertial frame; the velocity of P in this frame is constant, and the net external force on P is zero, so Newton's first law is obeyed in B . Observers in frames A and B will disagree about the velocity of P , but they will agree that P has a constant velocity (zero acceleration) and has zero net external force acting on it.

There is no single inertial frame of reference that is preferred over all others for formulating Newton's laws. If one frame is inertial, then every other frame moving relative to it with constant velocity is also inertial. Viewed in this light, the state of rest and the state of motion with constant velocity are not very different; both occur when the vector sum of forces acting on the object is zero.

TEST YOUR UNDERSTANDING OF SECTION 4.2 In which of the following situations is there zero net external force on the object? (i) An airplane flying due north at a steady 120 m/s and at a constant altitude; (ii) a car driving straight up a hill with a 3° slope at a constant 90 km/h; (iii) a hawk circling at a constant 20 km/h at a constant height of 15 m above an open field; (iv) a box with slick, frictionless surfaces in the back of a truck as the truck accelerates forward on a level road at 5 m/s^2 .

ANSWER In (i), the hawk is moving in a circle; hence it is accelerating and is *not* in equilibrium. In (iii), the hawk is moving in a circle; hence it is accelerating and is *not* in equilibrium. In (iv), the truck starts from rest, the box remains stationary as seen in the inertial reference frame of the ground as the truck accelerates forward. Like the person on skates in Fig. 4.10a, object is zero. [In (vi), if the truck starts from rest, the box remains stationary as seen in the inertial reference frame of the ground as the truck accelerates forward. Like the person on skates in Fig. 4.10a, object is zero. In (i), (ii), and (vi) the object is not accelerating, so the net external force on the object is zero. In (iii), the hawk is moving in a circle; hence it is accelerating and is *not* in equilibrium. In (iv), the truck starts from rest, the box remains stationary as seen in the inertial reference frame of the ground as the truck accelerates forward. Like the person on skates in Fig. 4.10a, object is zero.]

4.3 NEWTON'S SECOND LAW

Newton's first law tells us that when an object is acted on by zero net external force, the object moves with constant velocity and zero acceleration. In **Fig. 4.12a** (next page), a hockey puck is sliding to the right on wet ice. There is negligible friction, so there are no horizontal forces acting on the puck; the downward force of gravity and the upward normal force exerted by the ice surface sum to zero. So the net external force $\Sigma \vec{F}$ acting on the puck is zero, the puck has zero acceleration, and its velocity is constant.

But what happens when the net external force is *not* zero? In Fig. 4.12b we apply a constant horizontal force to a sliding puck in the same direction that the puck is moving. Then $\Sigma \vec{F}$ is constant and in the same horizontal direction as $\vec{v}$. We find that during the time the force is acting, the velocity of the puck changes at a constant rate; that is, the puck moves with constant acceleration. The speed of the puck increases, so the acceleration $\vec{a}$ is in the same direction as $\vec{v}$ and $\Sigma \vec{F}$.

In Fig. 4.12c we reverse the direction of the force on the puck so that $\Sigma \vec{F}$ acts opposite to $\vec{v}$. In this case as well, the puck has an acceleration; the puck moves more and more slowly to the right. The acceleration $\vec{a}$ in this case is to the left, in the same direction as $\Sigma \vec{F}$. As in the previous case, experiment shows that the acceleration is constant if $\Sigma \vec{F}$ is constant.

We conclude that *a net external force acting on an object causes the object to accelerate in the same direction as the net external force*. If the magnitude of the net external force is constant, as in Fig. 4.12b and Fig. 4.12c, then so is the magnitude of the acceleration.

Figure 4.11 From the frame of reference of the car, it seems as though a force is pushing the crash test dummies forward as the car comes to a sudden stop. But there is really no such force: As the car stops, the dummies keep moving forward as a consequence of Newton's first law.



Figure 4.12 Using a hockey puck on a frictionless surface to explore the relationship between the net external force $\Sigma \vec{F}$ on an object and the resulting acceleration $\vec{a}$ of the object.

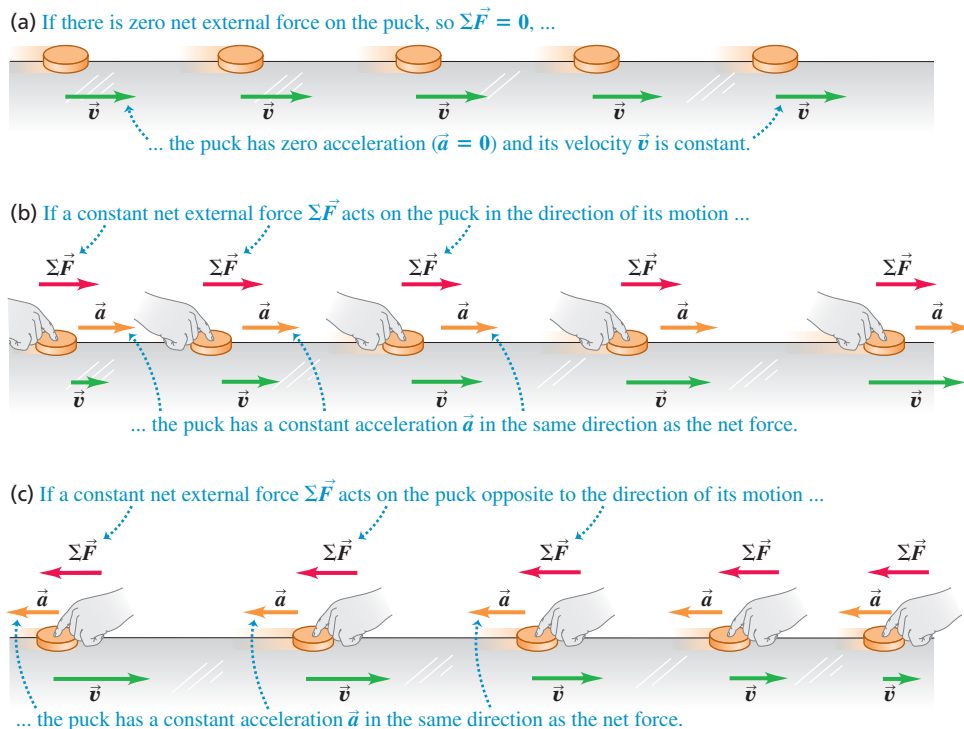


Figure 4.13 A top view of a hockey puck in uniform circular motion on a frictionless horizontal surface.

Puck moves at constant speed around circle.

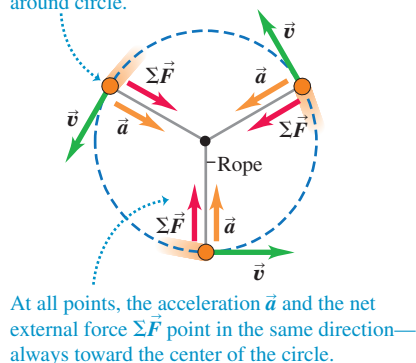
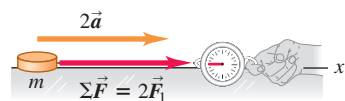


Figure 4.14 The magnitude of an object's acceleration $\vec{a}$ is directly proportional to the magnitude of the net external force $\Sigma \vec{F}$ acting on the object of mass m .

(a) A constant net external force $\Sigma \vec{F}$ causes a constant acceleration $\vec{a}$.



(b) Doubling the net external force doubles the acceleration.



(c) Halving the net external force halves the acceleration.



These conclusions about net external force and acceleration also apply to an object moving along a curved path. For example, **Fig. 4.13** shows a hockey puck moving in a horizontal circle on an ice surface of negligible friction. A rope is attached to the puck and to a stick in the ice, and this rope exerts an inward tension force of constant magnitude on the puck. The net external force and acceleration are both constant in magnitude and directed toward the center of the circle. The speed of the puck is constant, so this is uniform circular motion (see Section 3.4).

Figure 4.14a shows another experiment involving acceleration and net external force. We apply a constant horizontal force to a puck on a frictionless horizontal surface, using the spring balance described in Section 4.1 with the spring stretched a constant amount. As in Figs. 4.12b and Figs. 4.12c, this horizontal force equals the net external force on the puck. If we change the magnitude of the net external force, the acceleration changes in the same proportion. Doubling the net external force doubles the acceleration (Fig. 4.14b), halving the net external force halves the acceleration (Fig. 4.14c), and so on. Many such experiments show that *for any given object, the magnitude of the acceleration is directly proportional to the magnitude of the net external force acting on the object.*

Mass and Force

Our results mean that for a given object, the *ratio* of the magnitude $|\Sigma \vec{F}|$ of the net external force to the magnitude $a = |\vec{a}|$ of the acceleration is constant, regardless of the magnitude of the net external force. We call this ratio the *inertial mass*, or simply the **mass**, of the object and denote it by m . That is,

$$m = \frac{|\Sigma \vec{F}|}{a} \quad \text{or} \quad |\Sigma \vec{F}| = ma \quad \text{or} \quad a = \frac{|\Sigma \vec{F}|}{m} \quad (4.4)$$

Mass is a quantitative measure of inertia, which we discussed in Section 4.2. The last of the equations in Eqs. (4.4) says that the greater an object's mass, the more the object “resists” being accelerated. When you hold a piece of fruit in your hand at the supermarket and move it slightly up and down to estimate its heft, you’re applying a force and seeing how much the fruit accelerates up and down in response. If a force causes a large acceleration, the fruit has a small mass; if the same force causes only a small acceleration, the fruit has a large mass. In the same way, if you hit a table-tennis ball and then a basketball with

the same force, the basketball has much smaller acceleration because it has much greater mass.

The SI unit of mass is the **kilogram**. We mentioned in Section 1.3 that the kilogram is officially defined in terms of the definitions of the second and the meter, as well as the value of a fundamental quantity called Planck's constant. We can use this definition:

One newton is the amount of net external force that gives an acceleration of 1 meter per second squared to an object with a mass of 1 kilogram.

This definition allows us to calibrate the spring balances and other instruments used to measure forces. Because of the way we have defined the newton, it is related to the units of mass, length, and time. For Eqs. (4.4) to be dimensionally consistent, it must be true that

$$1 \text{ newton} = (1 \text{ kilogram})(1 \text{ meter per second squared})$$

or

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$

We'll use this relationship many times in the next few chapters, so keep it in mind.

Here's an application of Eqs. (4.4). Suppose we apply a constant net external force $\Sigma \vec{F}$ to an object of known mass m_1 and we find an acceleration of magnitude a_1 (Fig. 4.15a). We then apply the same force to another object of unknown mass m_2 , and we find an acceleration of magnitude a_2 (Fig. 4.15b). Then, according to Eqs. (4.4),

$$\begin{aligned} m_1 a_1 &= m_2 a_2 \\ \frac{m_2}{m_1} &= \frac{a_1}{a_2} \quad (\text{same net external force}) \end{aligned} \quad (4.5)$$

For the same net external force, the ratio of the masses of two objects is the inverse of the ratio of their accelerations. In principle we could use Eq. (4.5) to measure an unknown mass m_2 , but it is usually easier to determine mass indirectly by measuring the object's weight. We'll return to this point in Section 4.4.

When two objects with masses m_1 and m_2 are fastened together, we find that the mass of the composite object is always $m_1 + m_2$ (Fig. 4.15c). This additive property of mass may seem obvious, but it has to be verified experimentally. Ultimately, the mass of an object is related to the number of protons, electrons, and neutrons it contains. This wouldn't be a good way to *define* mass because there is no practical way to count these particles. But the concept of mass is the most fundamental way to characterize the quantity of matter in an object.

Stating Newton's Second Law

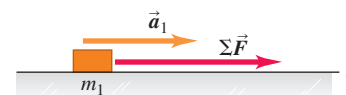
Experiment shows that the *net* external force on an object is what causes that object to accelerate. If a combination of forces $\vec{F}_1$, $\vec{F}_2$, $\vec{F}_3$, and so on is applied to an object, the object will have the same acceleration vector $\vec{a}$ as when only a single force is applied, if that single force is equal to the vector sum $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$. In other words, the principle of superposition of forces (see Fig. 4.4) also holds true when the net external force is not zero and the object is accelerating.

Equations (4.4) relate the magnitude of the net external force on an object to the magnitude of the acceleration that it produces. We have also seen that the direction of the net external force is the same as the direction of the acceleration, whether the object's path is straight or curved. Finally, we've seen that the forces that affect an object's motion are *external* forces, those exerted on the object by other objects in its environment. Newton wrapped up all these results into a single concise statement that we now call *Newton's second law of motion*:

NEWTON'S SECOND LAW OF MOTION If a net external force acts on an object, the object accelerates. The direction of acceleration is the same as the direction of the net external force. The mass of the object times the acceleration vector of the object equals the net external force vector.

Figure 4.15 For a given net external force $\Sigma \vec{F}$ acting on an object, the acceleration is inversely proportional to the mass of the object. Masses add like ordinary scalars.

(a) A known net external force $\Sigma \vec{F}$ causes an object with mass m_1 to have an acceleration $\vec{a}_1$.



(b) Applying the same net external force $\Sigma \vec{F}$ to a second object and noting the acceleration allow us to measure the mass.



(c) When the two objects are fastened together, the same method shows that their composite mass is the sum of their individual masses.

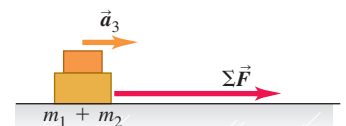


Figure 4.16 The design of high-performance motorcycles depends fundamentally on Newton's second law. To maximize the forward acceleration, the designer makes the motorcycle as light as possible (that is, minimizes the mass) and uses the most powerful engine possible (thus maximizing the forward force).



In symbols,

Newton's second law: If there is a net external force on an object ... $\Sigma \vec{F} = m\vec{a}$... the object accelerates in the same direction as the net external force.

Mass of object

(4.6)

An alternative statement is that the acceleration of an object is equal to the net external force acting on the object divided by the object's mass:

$$\vec{a} = \frac{\Sigma \vec{F}}{m}$$

Newton's second law is a fundamental law of nature, the basic relationship between force and motion. Most of the remainder of this chapter and all of the next are devoted to learning how to apply this principle in various situations.

Equation (4.6) has many practical applications (**Fig. 4.16**). You've actually been using it all your life to measure your body's acceleration. In your inner ear, microscopic hair cells are attached to a gelatinous substance that holds tiny crystals of calcium carbonate called *otoliths*. When your body accelerates, the hair cells pull the otoliths along with the rest of your body and sense the magnitude and direction of the force that they exert. By Newton's second law, the acceleration of the otoliths—and hence that of your body as a whole—is proportional to this force and has the same direction. In this way, you can sense the magnitude and direction of your acceleration even with your eyes closed!

Using Newton's Second Law

At least four aspects of Newton's second law deserve special attention. First, Eq. (4.6) is a *vector* equation. Usually we'll use it in component form, with a separate equation for each component of force and the corresponding component of acceleration:

Newton's second law: Each component of the net external force on an object ...

$$\Sigma F_x = ma_x \quad \Sigma F_y = ma_y \quad \Sigma F_z = ma_z$$

... equals the object's mass times the corresponding acceleration component.

(4.7)

This set of component equations is equivalent to the single vector Eq. (4.6).

Second, the statement of Newton's second law refers to *external* forces. As an example, how a kicked soccer ball moves isn't affected by the *internal* forces that hold the pieces of the ball together. That's why only external forces are included in the sum $\Sigma \vec{F}$ in Eqs. (4.6) and (4.7).

Third, Eqs. (4.6) and (4.7) are valid only when the mass m is *constant*. It's easy to think of systems whose masses change, such as a leaking tank truck or a moving railroad car being loaded with coal. Such systems are better handled by using the concept of momentum; we'll get to that in Chapter 8.

Finally, Newton's second law is valid in inertial frames of reference *only*, just like the first law. It's not valid in the reference frame of any of the accelerating vehicles in Fig. 4.10; relative to any of these frames, the passenger accelerates even though the net external force on the passenger is zero. We'll usually treat the earth as an adequate approximation to an inertial frame, although because of its rotation and orbital motion it is not precisely inertial.

CAUTION $m\vec{a}$ is **not** a force Even though the vector $m\vec{a}$ is equal to the vector sum $\Sigma \vec{F}$ of all the forces acting on the object, the vector $m\vec{a}$ is *not* a force. Acceleration is the *result* of the net external force; it is not a force itself. It's “common sense” to think that a “force of acceleration” pushes you back into your seat when your car accelerates forward from rest. But *there is no such force*; instead, your inertia causes you to tend to stay at rest relative to the earth, and the car accelerates around you (see Fig. 4.10a). The “common sense” confusion arises from trying to apply Newton's second law where it isn't valid—in the noninertial reference frame of an accelerating car. We'll always examine motion relative to *inertial* frames of reference only, and we strongly recommend that you do the same in solving problems. **I**

In learning how to use Newton's second law, we'll begin in this chapter with examples of straight-line motion. Then in Chapter 5 we'll consider more general kinds of motion and develop more detailed problem-solving strategies.

EXAMPLE 4.4 Newton's second law I: Determining acceleration from force

WITH **VAR**IATION PROBLEMS

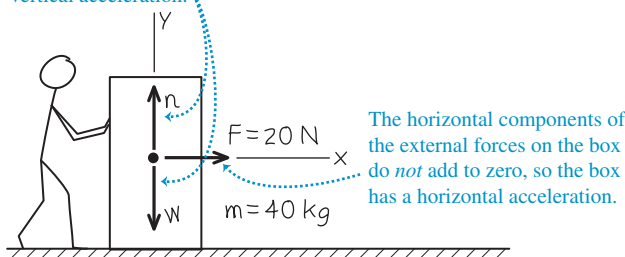
A worker applies a constant horizontal force with magnitude 20 N to a box with mass 40 kg resting on a level, freshly waxed floor with negligible friction. What is the acceleration of the box?

IDENTIFY and SET UP This problem involves force and acceleration, so we'll use Newton's second law. That means we'll have to find the net external force acting on the box and set it equal to the mass of the box multiplied by its acceleration. In this example, the acceleration is our target variable.

In *any* problem involving forces, to find the *net* external force we must first identify all of the *individual* external forces that act on the object in question. (Remember that the net external force is the vector sum of these individual forces.) To identify these forces, we'll use the idea that two broad categories of forces act on an object like the box in **Fig. 4.17**: the *weight* of the object $\vec{w}$ —that is, the downward gravitational force exerted by the earth—and *contact forces*, which are forces exerted by other objects that the object in question is touching. Two objects are touching the box—the worker's hands and the floor—and both exert contact forces on the box. The worker's hands exert a horizontal force $\vec{F}$ of magnitude 20 N. The floor exerts an upward supporting force; as in Section 4.1, we call this a *normal* force $\vec{n}$ because it acts perpendicular to the surface of contact. (Remember that “normal” is a synonym for “perpendicular.” It does not mean the opposite of “abnormal”!) If friction were present, the floor would also exert a friction force on the box; we'll ignore this here, since we're told that friction is negligible. Figure 4.17 shows these three external forces that act on the box.

Figure 4.17 Our sketch for this problem.

The vertical components of the external forces on the box sum to zero, and the box has no vertical acceleration.



Just as we did for the forces in Example 4.1 (Section 4.1), we'll find the vector sum of these external forces using components. That's why the second step in any problem involving forces is choosing a coordinate system for finding vector components. It's usually convenient to take one axis either along or opposite the direction of the object's acceleration, which in this case is horizontal. Hence we take the $+x$ -axis to be in the direction of the applied horizontal force (which is the direction in which the box accelerates) and the $+y$ -axis to be upward. In most force problems that you'll encounter (including this one), the force vectors all lie in a plane, so the z -axis isn't used.

The box doesn't move vertically, so the y -acceleration is zero: $a_y = 0$. Our target variable is the x -acceleration, a_x . We'll find it by using Newton's second law in component form, Eqs. (4.7).

EXECUTE The force $\vec{F}$ exerted by the worker has a positive x -component and zero y -component (so $F_x = F = 20$ N, $F_y = 0$); the normal force $\vec{n}$ has zero x -component and an upward, positive y -component (so $n_x = 0$, $n_y = n$); and the weight $\vec{w}$ has zero x -component and a downward, negative y -component (so $w_x = 0$, $w_y = -w$). From Newton's second law, Eqs. (4.7),

$$\Sigma F_x = F + 0 + 0 = F = 20 \text{ N} = ma_x$$

$$\Sigma F_y = 0 + n - w = ma_y = 0$$

From the first equation, the x -component of acceleration is

$$a_x = \frac{\Sigma F_x}{m} = \frac{20 \text{ N}}{40 \text{ kg}} = \frac{20 \text{ kg} \cdot \text{m/s}^2}{40 \text{ kg}} = 0.50 \text{ m/s}^2$$

EVALUATE The net external force is constant, so the acceleration in the $+x$ -direction is also constant. If we know the initial position and velocity of the box, we can find its position and velocity at any later time from the constant-acceleration equations of Chapter 2.

To determine a_x , we didn't need the y -component of Newton's second law from Eqs. (4.7), $\Sigma F_y = ma_y$. Can you use this equation to show that the magnitude n of the normal force in this situation is equal to the weight of the box?

KEYCONCEPT In problems involving forces and acceleration, first identify all of the external forces acting on an object, then choose a coordinate system. Find the vector sum of the external forces, and then set it equal to the mass of the object times the acceleration.

EXAMPLE 4.5 Newton's second law II: Determining force from acceleration

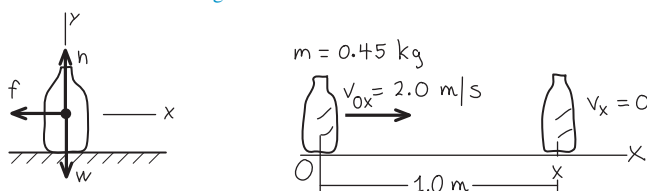
WITH **VAR**IATION PROBLEMS

A waitress shoves a ketchup bottle with mass 0.45 kg to her right along a smooth, level lunch counter. The bottle leaves her hand moving at 2.0 m/s, then slows down as it slides because of a constant horizontal friction force exerted on it by the countertop. It slides for 1.0 m before coming to rest. What are the magnitude and direction of the friction force acting on the bottle?

IDENTIFY and SET UP This problem involves forces and acceleration (the slowing of the ketchup bottle), so we'll use Newton's second law to solve it. As in Example 4.4, we identify the external forces acting on the bottle and choose a coordinate system (**Fig. 4.18**). And as in

Figure 4.18 Our sketch for this problem.

We draw one diagram showing the forces on the bottle and another one showing the bottle's motion.



Example 4.4, we have a downward gravitational force $\vec{w}$ and an upward normal force $\vec{n}$ exerted by the countertop. The countertop also exerts a friction force $\vec{f}$; this slows the bottle down, so its direction must be opposite the direction of the bottle's velocity (see Fig. 4.12c). We choose the $+x$ -axis to be in the direction that the bottle slides, and take the origin to be where the bottle leaves the waitress's hand.

Our target variable is the magnitude f of the friction force. We'll find it by using the x -component of Newton's second law from Eqs. (4.7). We aren't told the x -component of the bottle's acceleration, a_x , but we know that it's constant because the friction force that causes the acceleration is constant. Hence we can use a constant-acceleration formula from Section 2.4 to calculate a_x . We know the bottle's initial and final x -coordinates ($x_0 = 0$ and $x = 1.0$ m) and its initial and final x -velocity ($v_{0x} = 2.0$ m/s and $v_x = 0$), so the easiest equation to use is Eq. (2.13), $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$.

EXECUTE We solve Eq. (2.13) for a_x :

$$a_x = \frac{v_x^2 - v_{0x}^2}{2(x - x_0)} = \frac{(0 \text{ m/s})^2 - (2.0 \text{ m/s})^2}{2(1.0 \text{ m} - 0 \text{ m})} = -2.0 \text{ m/s}^2$$

The negative sign means that the bottle's acceleration is toward the *left* in Fig. 4.18, opposite to its velocity; this makes sense because the bottle is slowing down. As in Example 4.4, neither the normal force nor the weight has an x -component. That means the net external force in the x -direction is just the x -component $-f$ of the friction force:

$$\begin{aligned} \Sigma F_x &= -f = ma_x = (0.45 \text{ kg})(-2.0 \text{ m/s}^2) \\ &= -0.90 \text{ kg} \cdot \text{m/s}^2 = -0.90 \text{ N} \end{aligned}$$

The negative sign shows that the net external force on the bottle is toward the left. The *magnitude* of the friction force is $f = 0.90$ N.

EVALUATE As a check on the result, try repeating the calculation with the $+x$ -axis to the *left* in Fig. 4.18. You'll find that ΣF_x is equal to $+f = +0.90$ N (because the friction force is now in the $+x$ -direction), and again you'll find $f = 0.90$ N. The answers for the *magnitudes* of forces don't depend on the choice of coordinate axes!

We didn't write the y -component of Newton's second law in this example. You should do this and show that it's the same as in Example 4.4, so again the normal force and the weight have the same magnitude.

CAUTION Normal force and weight don't always have the same magnitude Be careful that you *never* assume automatically that the normal force $\vec{n}$ and the weight $\vec{w}$ have the same magnitude! Although that is the case in this example and the preceding one, we'll see many examples in Chapter 5 and later where the magnitude of the normal force is *not* equal to the weight. ■

KEYCONCEPT In problems involving forces in which you're given velocity, time, and/or displacement data, you'll need to use the equations for motion with constant acceleration as well as Newton's second law.

Figure 4.19 Despite its name, the British unit of mass has nothing to do with the type of slug shown here. A common garden slug has a mass of about 15 grams, or about 10^{-3} slug.



Some Notes on Units

A few words about units are in order. In the cgs metric system (not used in this book), the unit of mass is the gram, equal to 10^{-3} kg, and the unit of distance is the centimeter, equal to 10^{-2} m. The cgs unit of force is called the *dyne*:

$$1 \text{ dyne} = 1 \text{ g} \cdot \text{cm/s}^2 = 10^{-5} \text{ N}$$

In the British system, the unit of force is the *pound* (or pound-force) and the unit of mass is the *slug* (Fig. 4.19). The unit of acceleration is 1 foot per second squared, so

$$1 \text{ pound} = 1 \text{ slug} \cdot \text{ft/s}^2$$

The official definition of the pound is

$$1 \text{ pound} = 4.448221615260 \text{ newtons}$$

It is handy to remember that a pound is about 4.4 N and a newton is about 0.22 pound. Another useful fact: An object with a mass of 1 kg has a weight of about 2.2 lb at the earth's surface.

Table 4.2 lists the units of force, mass, and acceleration in the three systems.

TABLE 4.2 Units of Force, Mass, and Acceleration

System of Units	Force	Mass	Acceleration
SI	newton (N)	kilogram (kg)	m/s ²
cgs	dyne (dyn)	gram (g)	cm/s ²
British	pound (lb)	slug	ft/s ²

TEST YOUR UNDERSTANDING OF SECTION 4.3 Rank the following situations in order of the magnitude of the object's acceleration, from lowest to highest. Are there any cases that have the same magnitude of acceleration? (i) A 2.0 kg object acted on by a 2.0 N net force; (ii) a 2.0 kg object acted on by an 8.0 N net force; (iii) an 8.0 kg object acted on by a 2.0 N net force; (iv) an 8.0 kg object acted on by a 8.0 N net force.

ANSWER (i) $a = (2.0 \text{ N}) / (2.0 \text{ kg}) = 1.0 \text{ m/s}^2$; (ii) $a = (8.0 \text{ N}) / (2.0 \text{ kg}) = 4.0 \text{ m/s}^2$; (iii) $a = (2.0 \text{ N}) / (8.0 \text{ kg}) = 0.25 \text{ m/s}^2$; (iv) $a = (8.0 \text{ N}) / (8.0 \text{ kg}) = 1.0 \text{ m/s}^2$. The acceleration is equal to the net force divided by the mass. Hence (iii), (i) and (iv), (ii).

4.4 MASS AND WEIGHT

The *weight* of an object is the gravitational force that the earth exerts on the object. (If you are on another planet, your weight is the gravitational force that planet exerts on you.) Unfortunately, the terms “mass” and “weight” are often misused and interchanged in everyday conversation. It's absolutely essential for you to understand clearly the distinctions between these two physical quantities.

Mass characterizes the *inertial* properties of an object. Mass is what keeps the table setting on the table when you yank the tablecloth out from under it. The greater the mass, the greater the force needed to cause a given acceleration; this is reflected in Newton's second law, $\Sigma \vec{F} = m\vec{a}$.

Weight, on the other hand, is a *force* exerted on an object by the pull of the earth. Mass and weight are related: Objects that have large mass also have large weight. A large stone is hard to throw because of its large *mass*, and hard to lift off the ground because of its large *weight*.

To understand the relationship between mass and weight, note that a freely falling object has an acceleration of magnitude g (see Section 2.5). Newton's second law tells us that a force must act to produce this acceleration. If a 1 kg object falls with an acceleration of 9.8 m/s^2 , the required force has magnitude

$$F = ma = (1 \text{ kg})(9.8 \text{ m/s}^2) = 9.8 \text{ kg} \cdot \text{m/s}^2 = 9.8 \text{ N}$$

The force that makes the object accelerate downward is its weight. Any object near the surface of the earth that has a mass of 1 kg *must* have a weight of 9.8 N to give it the acceleration we observe when it is in free fall. More generally,

$$\text{Magnitude of weight of an object } w = mg \quad \text{Mass of object} \quad \text{Magnitude of acceleration due to gravity} \quad (4.8)$$

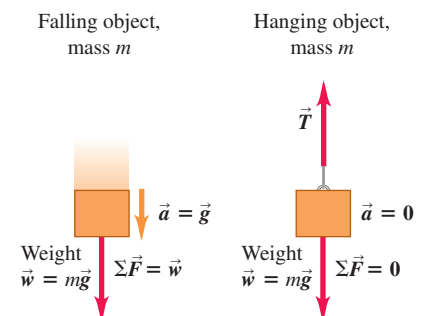
Hence the magnitude w of an object's weight is directly proportional to its mass m . The weight of an object is a force, a vector quantity, and we can write Eq. (4.8) as a vector equation (**Fig. 4.20**):

$$\vec{w} = m\vec{g} \quad (4.9)$$

Remember that g is the *magnitude* of $\vec{g}$, the acceleration due to gravity, so g is always a positive number, by definition. Thus w , given by Eq. (4.8), is the *magnitude* of the weight and is also always positive.

CAUTION An object's weight acts at all times When keeping track of the external forces on an object, remember that the weight is present *all the time*, whether the object is in free fall or not. If we suspend an object from a rope, it is in equilibrium and its acceleration is zero. But its weight, given by Eq. (4.9), is still pulling down on it (**Fig. 4.20**). In this case the rope pulls up on the object, applying an upward force. The *vector sum* of the external forces is zero, but the weight still acts. ■

Figure 4.20 Relating the mass and weight of an object.



- The relationship of mass to weight: $\vec{w} = m\vec{g}$.
- This relationship is the same whether an object is falling or stationary.

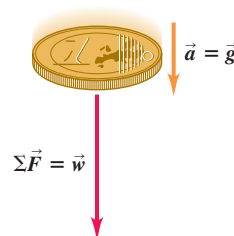
CONCEPTUAL EXAMPLE 4.6 Net external force and acceleration in free fall

In Example 2.6 of Section 2.5, a one-euro coin was dropped from rest from the Leaning Tower of Pisa. If the coin falls freely, so that the effects of the air are negligible, how does the net external force on the coin vary as it falls?

SOLUTION In free fall, the acceleration $\vec{a}$ of the coin is constant and equal to $\vec{g}$. Hence by Newton's second law the net external force $\Sigma\vec{F} = m\vec{a}$ is also constant and equal to $m\vec{g}$, which is the coin's weight $\vec{w}$ (Fig. 4.21). The coin's velocity changes as it falls, but the net external force acting on it is constant.

The net external force on a freely falling coin is constant even if you initially toss it upward. The force that your hand exerts on the coin to toss it is a contact force, and it disappears the instant the coin leaves your hand. From then on, the only force acting on the coin is its weight $\vec{w}$.

Figure 4.21 The acceleration of a freely falling object is constant, and so is the net external force acting on the object.



KEYCONCEPT The gravitational force on an object (its weight) does not depend on how the object is moving.

Figure 4.22 The weight of a 1 kilogram mass (a) on earth and (b) on the moon.

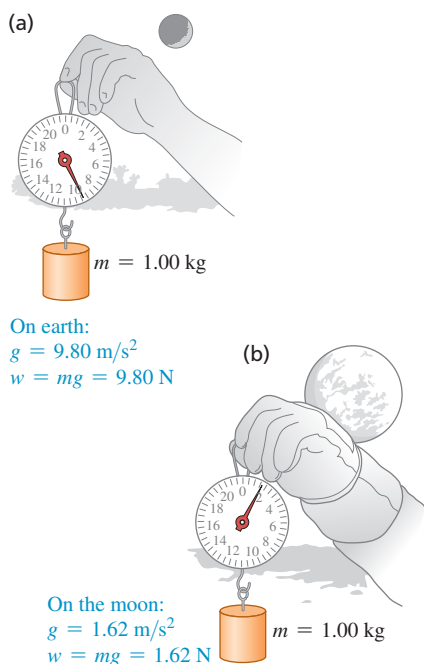
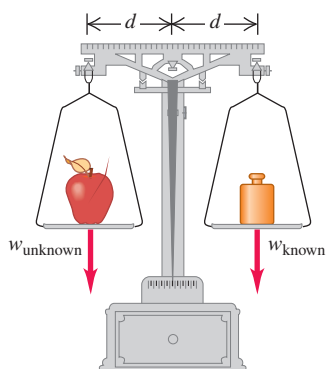


Figure 4.23 An equal-arm balance determines the mass of an object (such as an apple) by comparing its weight to a known weight.

**Variation of g with Location**

We'll use $g = 9.80 \text{ m/s}^2$ for problems set on the earth (or, if the other data in the problem are given to only two significant figures, $g = 9.8 \text{ m/s}^2$). In fact, the value of g varies somewhat from point to point on the earth's surface—from about 9.78 to 9.82 m/s^2 —because the earth is not perfectly spherical and because of effects due to its rotation. At a point where $g = 9.80 \text{ m/s}^2$, the weight of a standard kilogram is $w = 9.80 \text{ N}$. At a different point, where $g = 9.78 \text{ m/s}^2$, the weight is $w = 9.78 \text{ N}$ but the mass is still 1 kg . The weight of an object varies from one location to another; the mass does not.

If we take a standard kilogram to the surface of the moon, where the value of g is 1.62 m/s^2 , its weight is 1.62 N but its mass is still 1 kg (Fig. 4.22). An 80.0 kg astronaut has a weight on earth of $(80.0 \text{ kg})(9.80 \text{ m/s}^2) = 784 \text{ N}$, but on the moon the astronaut's weight would be only $(80.0 \text{ kg})(1.62 \text{ m/s}^2) = 130 \text{ N}$. In Chapter 13 we'll see how to calculate the value of g at the surface of the moon or on other worlds.

Measuring Mass and Weight

In Section 4.3 we described a way to compare masses by comparing their accelerations when they are subjected to the same net external force. Usually, however, the easiest way to measure the mass of an object is to measure its weight, often by comparing with a standard. Equation (4.8) says that two objects that have the same weight at a particular location also have the same mass. We can compare weights very precisely; the familiar equal-arm balance (Fig. 4.23) can determine with great precision (up to 1 part in 10^6) when the weights of two objects are equal and hence when their masses are equal.

The concept of mass plays two rather different roles in mechanics. The weight of an object (the gravitational force acting on it) is proportional to its mass as stated in the equation $w = mg$; we call the property related to gravitational interactions *gravitational mass*. On the other hand, we call the inertial property that appears in Newton's second law ($\Sigma\vec{F} = m\vec{a}$) the *inertial mass*. If these two quantities were different, the acceleration due to gravity might well be different for different objects. However, extraordinarily precise experiments have established that in fact the two *are* the same to a precision of better than one part in 10^{12} .

CAUTION **Don't confuse mass and weight** The SI units for mass and weight are often misused in everyday life. For example, it's incorrect to say "This box weighs 6 kg ." What this really means is that the *mass* of the box, probably determined indirectly by *weighing*, is 6 kg . Avoid this sloppy usage in your own work! In SI units, weight (a force) is measured in newtons, while mass is measured in kilograms. **I**

EXAMPLE 4.7 Mass and weight

A $2.45 \times 10^4 \text{ N}$ truck traveling in the $+x$ -direction makes an emergency stop; the x -component of the net external force acting on it is $-1.83 \times 10^4 \text{ N}$. What is its acceleration?

IDENTIFY and SET UP Our target variable is the x -component of the truck's acceleration, a_x . We use the x -component portion of Newton's second law, Eqs. (4.7), to relate force and acceleration. To do this, we need to know the truck's mass. The newton is a unit for force, however, so $2.49 \times 10^4 \text{ N}$ is the truck's *weight*, not its mass. Hence we'll first use Eq. (4.8) to determine the truck's mass from its weight. The truck has a positive x -velocity and is slowing down, so its x -acceleration will be negative.

EXECUTE The mass of the truck is

$$m = \frac{w}{g} = \frac{2.45 \times 10^4 \text{ N}}{9.80 \text{ m/s}^2} = \frac{2.45 \times 10^4 \text{ kg} \cdot \text{m/s}^2}{9.80 \text{ m/s}^2} = 2540 \text{ kg}$$

Then $\Sigma F_x = ma_x$ gives

$$a_x = \frac{\Sigma F_x}{m} = \frac{-1.83 \times 10^4 \text{ N}}{2540 \text{ kg}} = \frac{-1.83 \times 10^4 \text{ kg} \cdot \text{m/s}^2}{2540 \text{ kg}} = -7.20 \text{ m/s}^2$$

EVALUATE The negative sign means that the acceleration vector points in the negative x -direction, as we expected. The magnitude of this acceleration is pretty high; passengers in this truck will experience a lot of rearward force from their seat belts.

KEYCONCEPT In problems involving Newton's second law, make sure that for m you use the mass of the object, *not* its weight.

TEST YOUR UNDERSTANDING OF SECTION 4.4 Suppose an astronaut landed on a planet where $g = 19.6 \text{ m/s}^2$. Compared to earth, would it be easier, harder, or just as easy for her to walk around? Would it be easier, harder, or just as easy for her to catch a ball that is moving horizontally at 12 m/s ? (Assume that the astronaut's spacesuit is a lightweight model that doesn't impede her movements in any way.)

ANSWER

It would take twice the effort for the astronaut to walk around because her weight on the planet would be twice as much as on the earth. But it would be just as easy to catch a ball moving horizontally. The ball's *mass* is the same as on earth, so the horizontal force the astronaut would have to exert to bring it to a stop (i.e., to give it the same acceleration) would also be the same as on earth.

4.5 NEWTON'S THIRD LAW

A force acting on an object is always the result of its interaction with another object, so forces always come in pairs. You can't pull on a doorknob without the doorknob pulling back on you. When you kick a football, your foot exerts a forward force on the ball, but you also feel the force the ball exerts back on your foot.

In each of these cases, the force that you exert on the other object is in the opposite direction to the force that object exerts on you. Experiments show that whenever two objects interact, the two forces that they exert on each other are always *equal in magnitude* and *opposite in direction*. This fact is called *Newton's third law of motion*:

NEWTON'S THIRD LAW OF MOTION If object A exerts a force on object B (an “action”), then object B exerts a force on object A (a “reaction”). These two forces have the same magnitude but are opposite in direction. These two forces act on different objects.

For example, in **Fig. 4.24** $\vec{F}_{A \text{ on } B}$ is the force applied by object A (first subscript) on object B (second subscript), and $\vec{F}_{B \text{ on } A}$ is the force applied by object B (first subscript) on object A (second subscript). In equation form,

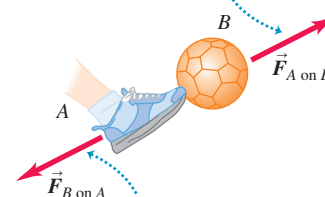
Newton's third law:
When two objects A and B exert forces on each other ...

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A} \quad \dots \text{the two forces have the same magnitude but opposite directions.} \quad (4.10)$$

Note: The two forces act on different objects.

Figure 4.24 Newton's third law of motion.

If object A exerts force $\vec{F}_{A \text{ on } B}$ on object B (for example, a foot kicks a ball) ...



... then object B necessarily exerts force $\vec{F}_{B \text{ on } A}$ on object A (ball kicks back on foot).

The two forces have the same magnitude but opposite directions: $\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$.

APPLICATION Blame Newton's

Laws This car stopped because of Newton's second and third laws. During the impact, the car exerted a force on the tree; in accordance with the third law, the tree exerted an equally strong force back on the car. In accordance with the second law, the force of the tree on the car gave the car an acceleration that changed its velocity to zero.



It doesn't matter whether one object is inanimate (like the soccer ball in Fig. 4.24) and the other is not (like the kicker's foot): They necessarily exert forces on each other that obey Eq. (4.10).

In the statement of Newton's third law, "action" and "reaction" are the two opposite forces (in Fig. 4.24, $\vec{F}_{A \text{ on } B}$ and $\vec{F}_{B \text{ on } A}$); we sometimes refer to them as an **action–reaction pair**. This is *not* meant to imply any cause-and-effect relationship; we can consider either force as the "action" and the other as the "reaction." We often say simply that the forces are "equal and opposite," meaning that they have equal magnitudes and opposite directions.

CAUTION **The two forces in an action–reaction pair act on different objects** We stress that the two forces described in Newton's third law act on *different* objects. This is important when you solve problems involving Newton's first or second law, which involve the forces that act *on* an object. For instance, the net external force on the soccer ball in Fig. 4.24 is the vector sum of the weight of the ball and the force $\vec{F}_{A \text{ on } B}$ exerted by kicker A on the ball B. You wouldn't include the force $\vec{F}_{B \text{ on } A}$ because this force acts on the kicker A, *not* on the ball. ■

In Fig. 4.24 the action and reaction forces are *contact* forces that are present only when the two objects are touching. But Newton's third law also applies to *long-range* forces that do not require physical contact, such as the force of gravitational attraction. A table-tennis ball exerts an upward gravitational force on the earth that's equal in magnitude to the downward gravitational force the earth exerts on the ball. When you drop the ball, both the ball and the earth accelerate toward each other. The net force on each object has the same magnitude, but the earth's acceleration is microscopically small because its mass is so great. Nevertheless, it does move!

CAUTION **Contact forces need contact** If your fingers push on an object, the force you exert acts only when your fingers and the object are in contact. Once contact is broken, the force is no longer present even if the object is still moving. ■

CONCEPTUAL EXAMPLE 4.8 Which force is greater?

After your sports car breaks down, you start to push it to the nearest repair shop. While the car is starting to move, how does the force you exert on the car compare to the force the car exerts on you? How do these forces compare when you are pushing the car along at a constant speed?

SOLUTION Newton's third law says that in *both* cases, the force you exert on the car is equal in magnitude and opposite in direction to the force the car exerts on you. It's true that you have to push harder to get the car going than to keep it going. But no matter how hard you push on the car, the car pushes just as hard back on you. Newton's third law gives the same result whether the two objects are at rest, moving with constant velocity, or accelerating.

You may wonder how the car "knows" to push back on you with the same magnitude of force that you exert on it. It may help to visualize the forces you and the car exert on each other as interactions between the atoms at the surface of your hand and the atoms at the surface of the car. These interactions are analogous to miniature springs between adjacent atoms, and a compressed spring exerts equally strong forces on both of its ends.

KEYCONCEPT No matter how two interacting objects are moving, the forces that they exert on each other always have the same magnitude and point in opposite directions.

CONCEPTUAL EXAMPLE 4.9 Newton's third law I: Objects at rest

An apple sits at rest on a table, in equilibrium. What forces act on the apple? What is the reaction force to each of the forces acting on the apple? What are the action–reaction pairs?

SOLUTION Figure 4.25a shows the forces acting on the apple. $\vec{F}_{\text{earth on apple}}$ is the weight of the apple—that is, the downward gravitational force exerted *by* the earth *on* the apple. Similarly, $\vec{F}_{\text{table on apple}}$ is the upward normal force exerted *by* the table *on* the apple.

Figure 4.25b shows one of the action–reaction pairs involving the apple. As the earth pulls down on the apple, with force $\vec{F}_{\text{earth on apple}}$, the

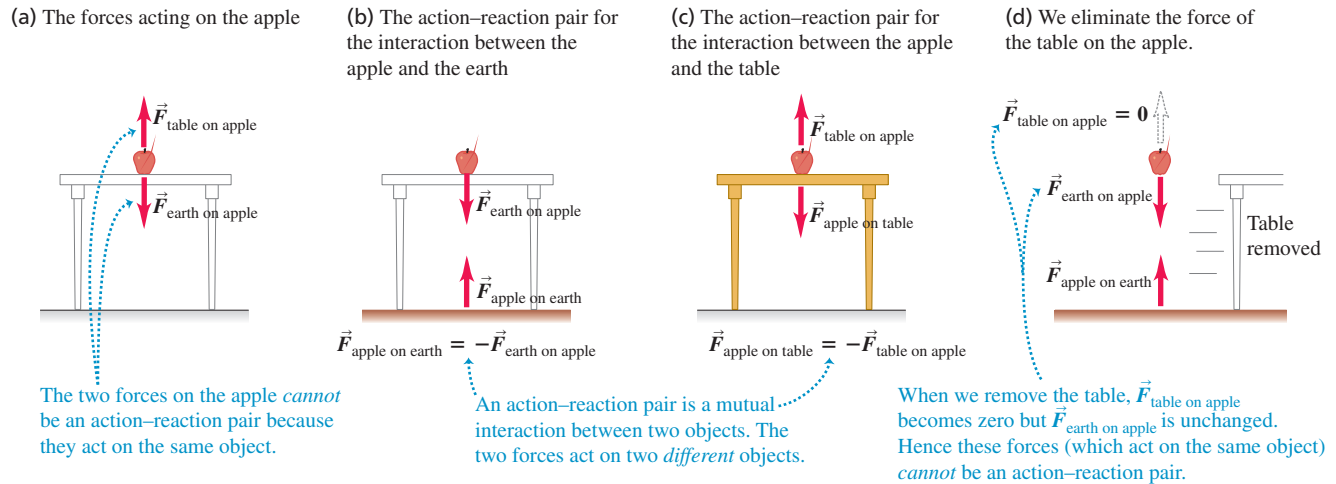
apple exerts an equally strong upward pull on the earth $\vec{F}_{\text{apple on earth}}$. By Newton's third law (Eq. 4.10) we have

$$\vec{F}_{\text{apple on earth}} = -\vec{F}_{\text{earth on apple}}$$

Figure 4.25c shows the other action–reaction pair involving the apple. The table pushes up on the apple with force $\vec{F}_{\text{table on apple}}$; the corresponding reaction is the downward force $\vec{F}_{\text{apple on table}}$ exerted by the apple on the table. For this action–reaction pair we have

$$\vec{F}_{\text{apple on table}} = -\vec{F}_{\text{table on apple}}$$

Figure 4.25 Identifying action–reaction pairs.



The two forces acting on the apple in Fig. 4.25a, $\vec{F}_{\text{table on apple}}$ and $\vec{F}_{\text{earth on apple}}$, are *not* an action–reaction pair, despite being equal in magnitude and opposite in direction. They do not represent the mutual interaction of two objects; they are two different forces acting on the *same* object. Figure 4.25d shows another way to see this. If we suddenly yank the table out from under the apple, the forces $\vec{F}_{\text{apple on table}}$ and $\vec{F}_{\text{table on apple}}$ suddenly become zero, but $\vec{F}_{\text{apple on earth}}$ and $\vec{F}_{\text{earth on apple}}$ are unchanged (the gravitational interaction is still

present). Because $\vec{F}_{\text{table on apple}}$ is now zero, it can't be the negative of the nonzero $\vec{F}_{\text{earth on apple}}$, and these two forces can't be an action–reaction pair. *The two forces in an action–reaction pair never act on the same object.*

KEYCONCEPT The two forces in an action–reaction pair always act on two different objects.

CONCEPTUAL EXAMPLE 4.10 Newton's third law II: Objects in motion

A stonemason drags a marble block across a floor by pulling on a rope attached to the block (Fig. 4.26a). The block is not necessarily in equilibrium. What are the forces that correspond to the interactions between the block, rope, and mason? What are the action–reaction pairs?

SOLUTION We'll use the subscripts B for the block, R for the rope, and M for the mason. In Fig. 4.26b $\vec{F}_{\text{M on R}}$ is the force exerted by the *mason* on the *rope*, and the corresponding reaction is the force $\vec{F}_{\text{R on M}}$ exerted by the *rope* on the *mason*. Similarly, $\vec{F}_{\text{R on B}}$ is the force exerted by the *rope* on the *block*, and the corresponding reaction is the force $\vec{F}_{\text{B on R}}$ exerted by the *block* on the *rope*. The forces in each action–reaction pair are equal and opposite:

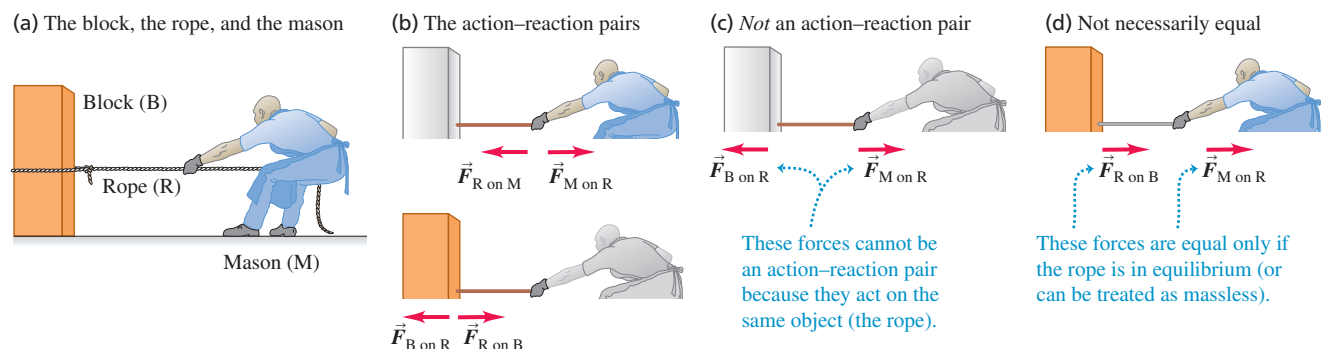
$$\vec{F}_{\text{R on M}} = -\vec{F}_{\text{M on R}} \quad \text{and} \quad \vec{F}_{\text{B on R}} = -\vec{F}_{\text{R on B}}$$

Forces $\vec{F}_{\text{M on R}}$ and $\vec{F}_{\text{B on R}}$ (Fig. 4.26c) are *not* an action–reaction pair because both of these forces act on the *same* object (the rope); an action and its reaction *must* always act on *different* objects. Furthermore, the forces $\vec{F}_{\text{M on R}}$ and $\vec{F}_{\text{B on R}}$ are not necessarily equal in magnitude. Applying Newton's second law to the rope, we get

$$\Sigma \vec{F} = \vec{F}_{\text{M on R}} + \vec{F}_{\text{B on R}} = m_{\text{R}} \vec{a}_{\text{R}}$$

If the block and rope are accelerating (speeding up or slowing down), the rope is *not* in equilibrium, and $\vec{F}_{\text{M on R}}$ must have a different magnitude than $\vec{F}_{\text{B on R}}$. By contrast, the action–reaction forces $\vec{F}_{\text{M on R}}$ and $\vec{F}_{\text{R on M}}$ are always equal in magnitude, as are $\vec{F}_{\text{R on B}}$ and $\vec{F}_{\text{B on R}}$. Newton's third law holds whether or not the objects are accelerating.

Figure 4.26 Identifying the interaction forces when a mason pulls on a rope attached to a block.



Continued

In the special case in which the rope is in equilibrium, the forces $\vec{F}_{M \text{ on } R}$ and $\vec{F}_{B \text{ on } R}$ are equal in magnitude and opposite in direction. But this is an example of Newton's *first* law, not his third; these are two forces on the same object, not forces of two objects on each other. Another way to look at this is that in equilibrium, $\vec{a}_R = 0$ in the previous equation. Then $\vec{F}_{B \text{ on } R} = -\vec{F}_{M \text{ on } R}$ because of Newton's first law.

Another special case is if the rope is accelerating but has negligibly small mass compared to that of the block or the mason. In this case, $m_R = 0$ in the previous equation, so again $\vec{F}_{B \text{ on } R} = -\vec{F}_{M \text{ on } R}$. Since Newton's third law says that $\vec{F}_{B \text{ on } R}$ *always* equals $-\vec{F}_{R \text{ on } B}$ (they are an action–reaction pair), in this “massless-rope” case $\vec{F}_{R \text{ on } B}$ also equals $\vec{F}_{M \text{ on } R}$.

CONCEPTUAL EXAMPLE 4.11 A Newton's third law paradox?

We saw in Conceptual Example 4.10 that the stonemason pulls as hard on the rope–block combination as that combination pulls back on him. Why, then, does the block move while the stonemason remains stationary?

SOLUTION To resolve this seeming paradox, keep in mind the difference between Newton's *second* and *third* laws. The only forces involved in Newton's second law are those that act *on* a given object. The vector sum of these forces determines the object's acceleration, if any. By contrast, Newton's third law relates the forces that two *different* objects exert on *each other*. The third law alone tells you nothing about the motion of either object.

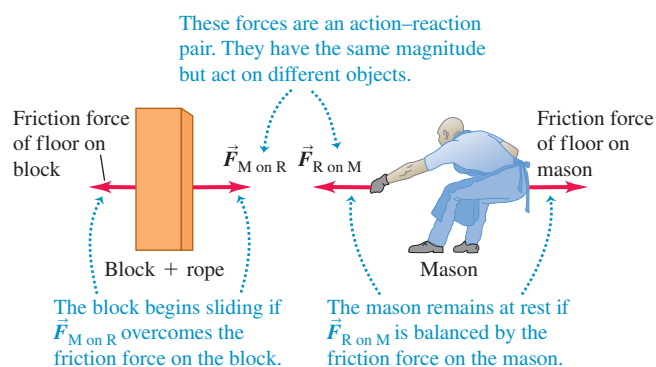
If the rope–block combination is initially at rest, it begins to slide if the stonemason exerts a force on the rope $\vec{F}_{M \text{ on } R}$ that is *greater* in magnitude than the friction force that the floor exerts on the block (Fig. 4.27). Then there is a net external force to the right on the rope–block combination, and it accelerates to the right. By contrast, the stonemason *doesn't* move because the net external force acting on him is *zero*. His shoes have nonskid soles that don't slip on the floor, so the friction force that the floor exerts on him is strong enough to balance the pull of the rope on him, $\vec{F}_{R \text{ on } M}$. (Both the block and the stonemason also experience a downward force of gravity and an upward normal force exerted by the floor. These forces balance each other, and so don't affect the motion of the block or the mason.)

Once the block is moving, the stonemason doesn't need to pull as hard; he must exert only enough force to balance the friction force on the block. Then the net external force on the moving block is zero, and by Newton's first law the block continues to move toward the mason at a constant velocity.

For both the “massless-rope” case and a rope in equilibrium, the rope exerts the same force on the block as the mason exerts on the rope (Fig. 4.26d). Hence we can think of the rope as “transmitting” to the block the force the mason exerts on the rope. But remember that this is true *only* when the rope has negligibly small mass or is in equilibrium.

KEYCONCEPT In problems that involve more than one object, use Newton's third law to relate the forces that the objects exert on each other.

Figure 4.27 The horizontal forces acting on the block–rope combination (left) and the mason (right). (The vertical forces are not shown.)



So the block accelerates but the stonemason doesn't because different amounts of friction act on them. If the floor were freshly waxed, so that there was little friction between the floor and the stonemason's shoes, pulling on the rope might start the block sliding to the right *and* start him sliding to the left.

Here's the moral: When analyzing the motion of an object, remember that only the forces acting *on* an object determine its motion. From this perspective, Newton's third law is merely a tool that can help you determine what those forces are.

KEYCONCEPT The motion of an object depends on the forces that are exerted on it, not the forces that it exerts on other objects.

An object that has pulling forces applied at its ends, such as the rope in Fig. 4.26, is said to be in *tension*. The **tension** at any point along the rope is the magnitude of the force acting at that point (see Fig. 4.2c). In Fig. 4.26b the tension at the right end of the rope is the magnitude of $\vec{F}_{M \text{ on } R}$ (or of $\vec{F}_{R \text{ on } M}$), and the tension at the left end is the magnitude of $\vec{F}_{B \text{ on } R}$ (or of $\vec{F}_{R \text{ on } B}$). If the rope is in equilibrium and if no forces act except at its ends, the net external force on the rope is zero and the tension is the *same* at both ends and throughout the rope. Thus, if the magnitudes of $\vec{F}_{B \text{ on } R}$ and $\vec{F}_{M \text{ on } R}$ are 50 N each, the tension in the rope is 50 N (*not* 100 N). The same is true if we can regard the rope as “massless” (that is, if its mass is small compared to that of the objects to which it's attached).

We emphasize once again that the two forces in an action–reaction pair *never* act on the same object. Remembering this fact can help you avoid confusion about action–reaction pairs and Newton's third law.

TEST YOUR UNDERSTANDING OF SECTION 4.5 You are driving a car on a country road when a mosquito splatters on the windshield. Which has the greater magnitude: the force that the car exerted on the mosquito or the force that the mosquito exerted on the car? Or are the magnitudes the same? If they are different, how can you reconcile this fact with Newton’s third law? If they are equal, why is the mosquito splattered while the car is undamaged?

ANSWER

By Newton’s third law, the two forces have equal magnitude. Because the car has much greater mass than the mosquito, it undergoes only a tiny, imperceptible acceleration in response to the force of the impact. By contrast, the mosquito, with its minuscule mass, undergoes a catastrophically large acceleration.

4.6 FREE-BODY DIAGRAMS

Newton’s three laws of motion contain all the basic principles we need to solve a wide variety of problems in mechanics. These laws are very simple in form, but the process of applying them to specific situations can pose real challenges. In this brief section we’ll point out three key ideas and techniques to use in any problems involving Newton’s laws. You’ll learn others in Chapter 5, which also extends the use of Newton’s laws to cover more complex situations.

1. *Newton’s first and second laws apply to a specific object.* Whenever you use Newton’s first law, $\Sigma \vec{F} = \mathbf{0}$, for an equilibrium situation or Newton’s second law, $\Sigma \vec{F} = m\vec{a}$, for a nonequilibrium situation, you must decide at the beginning to which object you are referring. This decision may sound trivial, but it isn’t.
2. *Only forces acting on the object matter.* The sum $\Sigma \vec{F}$ includes all the forces that act *on* the object in question. Hence, once you’ve chosen the object to analyze, you have to identify all the forces acting on it. Don’t confuse the forces acting on a object with the forces exerted by that object on some other object. For example, to analyze a person walking, you would include in $\Sigma \vec{F}$ the force that the ground exerts on the person as he walks, but *not* the force that the person exerts on the ground (**Fig. 4.28**). These forces form an action–reaction pair and are related by Newton’s third law, but only the member of the pair that acts on the object you’re working with goes into $\Sigma \vec{F}$.
3. *Free-body diagrams are essential to help identify the relevant forces.* A **free-body diagram** shows the chosen object by itself, “free” of its surroundings, with vectors drawn to show the magnitudes and directions of all the forces that act on the object. (Here “body” is another word for “object.”) We’ve shown free-body diagrams in Figs. 4.17, 4.18, 4.20, and 4.25a. Be careful to include all the forces acting *on* the object, but be equally careful *not* to include any forces that the object exerts on any other object. In particular, the two forces in an action–reaction pair must *never* appear in the same free-body diagram because they never act on the same object. Furthermore, never include forces that a object exerts on itself, since these can’t affect the object’s motion.

When a problem involves more than one object, you have to take the problem apart and draw a separate free-body diagram for each object. For example, Fig. 4.26c shows a separate free-body diagram for the rope in the case in which the rope is considered massless (so that no gravitational force acts on it). Figure 4.27 also shows diagrams for the block and the mason, but these are *not* complete free-body diagrams because they don’t show all the forces acting on each object. (We left out the vertical forces—the weight force exerted by the earth and the upward normal force exerted by the floor.)

In **Fig. 4.29** (next page) we present three real-life situations and the corresponding complete free-body diagrams. Note that in each situation a person exerts a force on something in his or her surroundings, but the force that shows up in the person’s free-body diagram is the surroundings pushing back *on* the person.

CAUTION Forces in free-body diagrams For a free-body diagram to be complete, you *must* be able to answer this question for each force: What other object is applying this force? If you can’t answer that question, you may be dealing with a nonexistent force. Avoid nonexistent forces such as “the force of acceleration” or “the $m\vec{a}$ force,” discussed in Section 4.3. **|**

Figure 4.28 The simple act of walking depends crucially on Newton’s third law. To start moving forward, you push backward on the ground with your foot. As a reaction, the ground pushes forward on your foot (and hence on your body as a whole) with a force of the same magnitude. This *external* force provided by the ground is what accelerates your body forward.

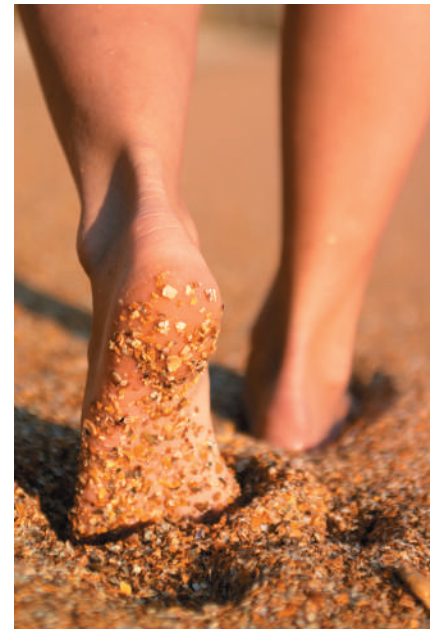
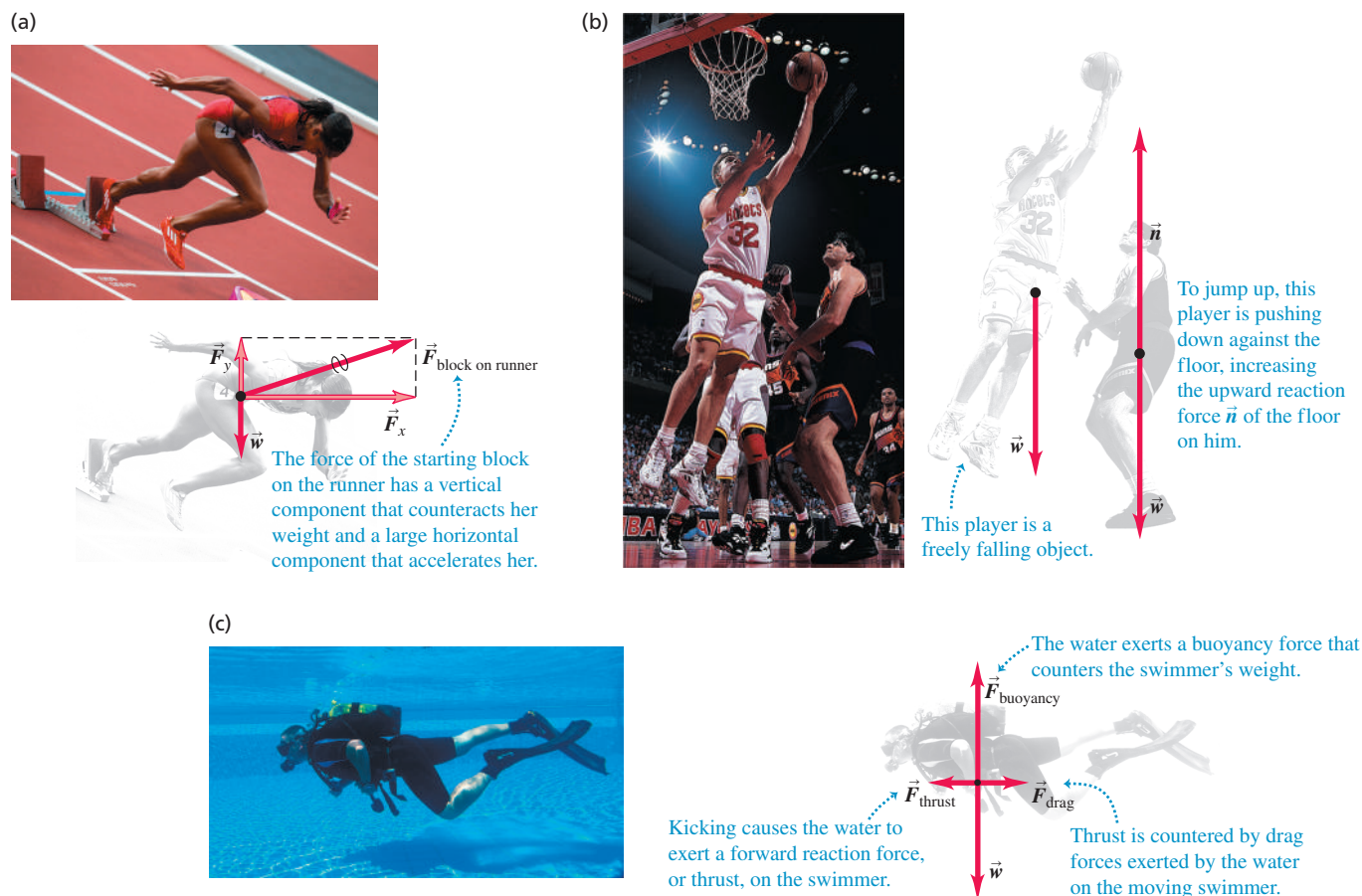


Figure 4.29 Examples of free-body diagrams. Each free-body diagram shows all of the external forces that act on the object in question.



TEST YOUR UNDERSTANDING OF SECTION 4.6 The buoyancy force shown in Fig. 4.29c is one half of an action–reaction pair. What force is the other half of this pair? (i) The weight of the swimmer; (ii) the forward thrust force; (iii) the backward drag force; (iv) the downward force that the swimmer exerts on the water; (v) the backward force that the swimmer exerts on the water by kicking.

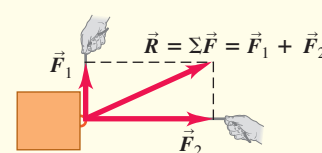
ANSWER

(iv) The buoyancy force is an *upward* force that the *water* exerts on the *swimmer*. By Newton's third law, the other half of the action–reaction pair is a *downward* force that the *swimmer* exerts on the *water* and has the same magnitude as the buoyancy force. It's true that the weight of the swimmer is also downward and has the same magnitude as the buoyancy force; however, the weight acts on the same object (the swimmer) as the buoyancy force, and so these forces aren't an action–reaction pair.

CHAPTER 4 SUMMARY

Force as a vector: Force is a quantitative measure of the interaction between two objects. It is a vector quantity. When several external forces act on an object, the effect on its motion is the same as if a single force, equal to the vector sum (resultant) of the forces, acts on the object. (See Example 4.1.)

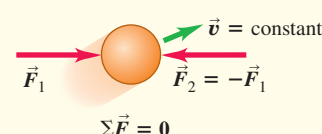
$$\vec{R} = \Sigma \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \cdots \quad (4.1)$$



The net external force on an object and Newton's first law:

Newton's first law states that when the vector sum of all external forces acting on a object (the *net external force*) is zero, the object is in equilibrium and has zero acceleration. If the object is initially at rest, it remains at rest; if it is initially in motion, it continues to move with constant velocity. This law is valid in inertial frames of reference only. (See Examples 4.2 and 4.3.)

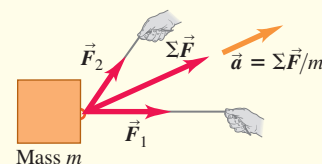
$$\Sigma \vec{F} = 0 \quad (4.3)$$



Mass, acceleration, and Newton's second law: The inertial properties of an object are characterized by its *mass*. Newton's second law states that the acceleration of an object under the action of a given set of external forces is directly proportional to the vector sum of the forces (the *net force*) and inversely proportional to the mass of the object. Like Newton's first law, this law is valid in inertial frames of reference only. In SI units, the unit of force is the newton (N), equal to $1 \text{ kg} \cdot \text{m/s}^2$. (See Examples 4.4 and 4.5.)

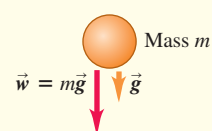
$$\Sigma \vec{F} = m\vec{a} \quad (4.6)$$

$$\begin{aligned} \Sigma F_x &= ma_x \\ \Sigma F_y &= ma_y \\ \Sigma F_z &= ma_z \end{aligned} \quad (4.7)$$



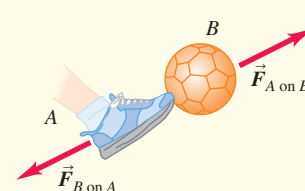
Weight: The weight $\vec{w}$ of an object is the gravitational force exerted on it by the earth. Weight is a vector quantity. The magnitude of the weight of an object at any specific location is equal to the product of its mass m and the magnitude of the acceleration due to gravity g at that location. The weight of an object depends on its location; its mass does not. (See Examples 4.6 and 4.7.)

$$w = mg \quad (4.8)$$



Newton's third law and action–reaction pairs: Newton's third law states that when two objects interact, they exert forces on each other that are equal in magnitude and opposite in direction. These forces are called action and reaction forces. Each of these two forces acts on only one of the two objects; they never act on the same object. (See Examples 4.8–4.11.)

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A} \quad (4.10)$$



Chapter 4 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 4.1** (Section 4.1) before attempting these problems.

VP4.1.1 Three professional wrestlers are fighting over a champion's belt, and each exerts a force on the belt. Wrestler 1 exerts a force $F_1 = 40.0 \text{ N}$ in the $+x$ -direction, wrestler 2 exerts a force $F_2 = 80.0 \text{ N}$ in the $-y$ -direction, and wrestler 3 exerts a force $F_3 = 60.0 \text{ N}$ at an

angle of 36.9° counterclockwise from the $+x$ -direction. Find the x - and y -components of the net external force on the belt, and find the force's magnitude and direction.

VP4.1.2 Three forces act on a statue. Force $\vec{F}_1$ (magnitude 45.0 N) points in the $+x$ -direction, force $\vec{F}_2$ (magnitude 105 N) points in the $+y$ -direction, and force $\vec{F}_3$ (magnitude 235 N) is at an angle of 36.9° from the $-x$ -direction and 53.1° from the $+y$ -direction. Find the x - and y -components of the net external force on the statue, and find the force's magnitude and direction.

VP4.1.3 An eagle descends steeply onto its prey. Its weight (the gravitational force on the eagle), of magnitude 60.0 N, points downward in the $-y$ -direction. The lift force exerted on the eagle's wings by the air, also of magnitude 60.0 N, is at an angle of 20.0° from the vertical (the $+y$ -direction) and 70.0° from the $+x$ -direction. The drag force (air resistance) exerted on the eagle by the air has magnitude 15.0 N and is at an angle of 20.0° from the $-x$ -direction and 70.0° from the $+y$ -direction. Find the x - and y -components of the net external force on the eagle, and find the force's magnitude and direction.

VP4.1.4 A box containing pizza sits on a table. Ernesto, who sits due east of the pizza box, pulls the box toward him with a force of 35.0 N. Kamala, who sits due north of the pizza box, pulls the box toward her with a 50.0 N force. Tsureku also sits at the table and pulls the box toward her so that the net external force on the box is 24.0 N in a direction 30.0° south of west. Take the $+x$ -direction to be due east and the $+y$ -direction to be due north. Find the x - and y -components of the force that Tsureku exerts, and find the force's magnitude and direction.

Be sure to review EXAMPLE 4.4 (Section 4.3) before attempting these problems.

VP4.4.1 A box of books with mass 55 kg rests on the level floor of the campus bookstore. The floor is freshly waxed and has negligible friction. A bookstore worker applies a constant horizontal force with magnitude 25 N to the box. What is the magnitude of the acceleration of the box?

VP4.4.2 A block of cheese of mass 2.0 kg sits on a freshly waxed, essentially frictionless table. You apply a constant horizontal force of 0.50 N to the cheese. (a) Name the three external forces that act on the cheese and what exerts each force. (b) What is the magnitude of the acceleration of the cheese?

VP4.4.3 In a game of ice hockey, you use a hockey stick to hit a puck of mass 0.16 kg that slides on essentially frictionless ice. During the hit you exert a constant horizontal force on the puck that gives it an acceleration of 75 m/s^2 for a fraction of a second. (a) During the hit, what is the magnitude of the horizontal force that you exert on the puck? (b) How does the magnitude of the normal force due to the ice compare to the weight of the puck?

VP4.4.4 A plate of cafeteria food is on a horizontal table. You push it away from you with a constant horizontal force of 14.0 N. The plate has a mass of 0.800 kg, and during the push it has an acceleration of 12.0 m/s^2 in the direction you are pushing it. (a) What is the magnitude of the net external force on the plate during the push? (b) What are the magnitude and direction of the friction force that the table exerts on the plate during the push?

Be sure to review EXAMPLE 4.5 (Section 4.3) before attempting these problems.

VP4.5.1 On a winter day a child of mass 20.0 kg slides on a horizontal sidewalk covered in ice. Initially she is moving at 3.00 m/s, but due to friction she comes to a halt in 2.25 m. What is the magnitude of the constant friction force that acts on her as she slides?

VP4.5.2 An airliner of mass $1.70 \times 10^5 \text{ kg}$ lands at a speed of 75.0 m/s. As it travels along the runway, the combined effects of air resistance, friction from the tires, and reverse thrust from the engines produce a constant force of $2.90 \times 10^5 \text{ N}$ opposite to the airliner's motion. What distance along the runway does the airliner travel before coming to a halt?

VP4.5.3 A truck of mass $2.40 \times 10^3 \text{ kg}$ is moving at 25.0 m/s. When the driver applies the brakes, the truck comes a stop after traveling 48.0 m. (a) How much time is required for the truck to stop? (b) What is the magnitude of the truck's constant acceleration as it slows down? (c) What is the magnitude of the constant braking force that acts on the truck as it slows down?

VP4.5.4 A car of mass $1.15 \times 10^3 \text{ kg}$ is stalled on a horizontal road. You and your friends give the car a constant, forward, horizontal push. There is friction between the car and the road. (a) Name the four external forces that act on the car as you and your friends push it and what exerts each force. (You can regard the combined push from you and your friends as a single force.) (b) The combined force that you and your friends exert has magnitude $8.00 \times 10^2 \text{ N}$, and starting from rest the car reaches a speed of 1.40 m/s after you have pushed it 5.00 m. Find the magnitude of the constant friction force that acts on the car.

BRIDGING PROBLEM Links in a Chain

A student suspends a chain consisting of three links, each of mass $m = 0.250 \text{ kg}$, from a light rope. The rope is attached to the top link of the chain, which does not swing. She pulls upward on the rope, so that the rope applies an upward force of 9.00 N to the chain. (a) Draw a free-body diagram for the entire chain, considered as an object, and one for each of the three links. (b) Use the diagrams of part (a) and Newton's laws to find (i) the acceleration of the chain, (ii) the force exerted by the top link on the middle link, and (iii) the force exerted by the middle link on the bottom link. Treat the rope as massless.

SOLUTION GUIDE

IDENTIFY and SET UP

1. There are four objects of interest in this problem: the chain as a whole and the three individual links. For each of these four objects, make a list of the external forces that act on it. Besides the force of gravity, your list should include only forces exerted by other objects that *touch* the object in question.
2. Some of the forces in your lists form action–reaction pairs (one pair is the force of the top link on the middle link and the force of the middle link on the top link). Identify all such pairs.
3. Use your lists to help you draw a free-body diagram for each of the four objects. Choose the coordinate axes.

4. Use your lists to decide how many unknowns there are in this problem. Which of these are target variables?

EXECUTE

5. Write a Newton's second law equation for each of the four objects, and write a Newton's third law equation for each action–reaction pair. You should have at least as many equations as there are unknowns (see step 4). Do you?
6. Solve the equations for the target variables.

EVALUATE

7. You can check your results by substituting them back into the equations from step 5. This is especially important to do if you ended up with more equations in step 5 than you used in step 6.
8. Rank the force of the rope on the chain, the force of the top link on the middle link, and the force of the middle link on the bottom link in order from smallest to largest magnitude. Does this ranking make sense? Explain.
9. Repeat the problem for the case in which the upward force that the rope exerts on the chain is only 7.35 N. Is the ranking in step 8 the same? Does this make sense?

PROBLEMS

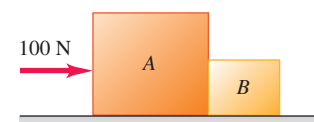
•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

- Q4.1** Can an object be in equilibrium when only one force acts on it? Explain.
- Q4.2** A ball thrown straight up has zero velocity at its highest point. Is the ball in equilibrium at this point? Why or why not?
- Q4.3** A helium balloon hovers in midair, neither ascending nor descending. Is it in equilibrium? What forces act on it?
- Q4.4** When you fly in an airplane at night in smooth air, you have no sensation of motion, even though the plane may be moving at 800 km/h (500 mi/h). Why?
- Q4.5** If the two ends of a rope in equilibrium are pulled with forces of equal magnitude and opposite directions, why isn't the total tension in the rope zero?
- Q4.6** You tie a brick to the end of a rope and whirl the brick around you in a horizontal circle. Describe the path of the brick after you suddenly let go of the rope.
- Q4.7** When a car stops suddenly, the passengers tend to move forward relative to their seats. Why? When a car makes a sharp turn, the passengers tend to slide to one side of the car. Why?
- Q4.8** Some people say that the “force of inertia” (or “force of momentum”) throws the passengers forward when a car brakes sharply. What is wrong with this explanation?
- Q4.9** A passenger in a moving bus with no windows notices that a ball that has been at rest in the aisle suddenly starts to move toward the rear of the bus. Think of two possible explanations, and devise a way to decide which is correct.
- Q4.10** Suppose you chose the fundamental physical quantities to be force, length, and time instead of mass, length, and time. What would be the units of mass in terms of those fundamental quantities?
- Q4.11** Why is the earth only approximately an inertial reference frame?
- Q4.12** Does Newton's second law hold true for an observer in a van as it speeds up, slows down, or rounds a corner? Explain.
- Q4.13** Some students refer to the quantity $m\vec{a}$ as “the force of acceleration.” Is it correct to refer to this quantity as a force? If so, what exerts this force? If not, what is a better description of this quantity?
- Q4.14** The acceleration of a falling object is measured in an elevator that is traveling upward at a constant speed of 9.8 m/s. What value is obtained?
- Q4.15** You can play catch with a softball in a bus moving with constant speed on a straight road, just as though the bus were at rest. Is this still possible when the bus is making a turn at constant speed on a level road? Why or why not?
- Q4.16** Students sometimes say that the force of gravity on an object is 9.8 m/s^2 . What is wrong with this view?
- Q4.17** Why can it hurt your foot more to kick a big rock than a small pebble? *Must* the big rock hurt more? Explain.
- Q4.18** “It's not the fall that hurts you; it's the sudden stop at the bottom.” Translate this saying into the language of Newton's laws of motion.
- Q4.19** A person can dive into water from a height of 10 m without injury, but a person who jumps off the roof of a 10-m-tall building and lands on a concrete street is likely to be seriously injured. Why is there a difference?
- Q4.20** Why are cars designed to crumple in front and back for safety? Why not for side collisions and rollovers?
- Q4.21** When a string barely strong enough lifts a heavy weight, it can lift the weight by a steady pull; but if you jerk the string, it will break. Explain in terms of Newton's laws of motion.

- Q4.22** A large crate is suspended from the end of a vertical rope. Is the tension in the rope greater when the crate is at rest or when it is moving upward at constant speed? If the crate is traveling upward, is the tension in the rope greater when the crate is speeding up or when it is slowing down? In each case, explain in terms of Newton's laws of motion.
- Q4.23** Which feels a greater pull due to the earth's gravity: a 10 kg stone or a 20 kg stone? If you drop the two stones, why doesn't the 20 kg stone fall with twice the acceleration of the 10 kg stone? Explain.
- Q4.24** Why is it incorrect to say that 1.0 kg *equals* 2.2 lb?
- Q4.25** A horse is hitched to a wagon. Since the wagon pulls back on the horse just as hard as the horse pulls on the wagon, why doesn't the wagon remain in equilibrium, no matter how hard the horse pulls?
- Q4.26** True or false? You exert a push P on an object and it pushes back on you with a force F . If the object is moving at constant velocity, then F is equal to P , but if the object is being accelerated, then P must be greater than F .
- Q4.27** A large truck and a small compact car have a head-on collision. During the collision, the truck exerts a force $\vec{F}_{T \text{ on } C}$ on the car, and the car exerts a force $\vec{F}_{C \text{ on } T}$ on the truck. Which force has the larger magnitude, or are they the same? Does your answer depend on how fast each vehicle was moving before the collision? Why or why not?
- Q4.28** When a car comes to a stop on a level highway, what force causes it to slow down? When the car increases its speed on the same highway, what force causes it to speed up? Explain.
- Q4.29** A small compact car is pushing a large van that has broken down, and they travel along the road with equal velocities and accelerations. While the car is speeding up, is the force it exerts on the van larger than, smaller than, or the same magnitude as the force the van exerts on it? Which vehicle has the larger net force on it, or are the net forces the same? Explain.
- Q4.30** Consider a tug-of-war between two people who pull in opposite directions on the ends of a rope. By Newton's third law, the force that A exerts on B is just as great as the force that B exerts on A . So what determines who wins? (*Hint*: Draw a free-body diagram showing all the forces that act on each person.)
- Q4.31** Boxes A and B are in contact on a horizontal, frictionless surface. You push on box A with a horizontal 100 N force (**Fig. Q4.31**). Box A weighs 150 N, and box B weighs 50 N. Is the force that box A exerts on box B equal to 100 N, greater than 100 N, or less than 100 N? Explain.
- Q4.32** A manual for student pilots contains this passage: “When an airplane flies at a steady altitude, neither climbing nor descending, the upward lift force from the wings equals the plane's weight. When the plane is climbing at a steady rate, the upward lift is greater than the weight; when the plane is descending at a steady rate, the upward lift is less than the weight.” Are these statements correct? Explain.
- Q4.33** If your hands are wet and no towel is handy, you can remove some of the excess water by shaking them. Why does this work?
- Q4.34** If you squat down (such as when you examine the books on a bottom shelf) and then suddenly get up, you may temporarily feel light-headed. What do Newton's laws of motion have to say about why this happens?
- Q4.35** When a car is hit from behind, the occupants may experience whiplash. Use Newton's laws of motion to explain what causes this result.
- Q4.36** In a head-on auto collision, passengers who are not wearing seat belts may be thrown through the windshield. Use Newton's laws of motion to explain why this happens.

Figure Q4.31



Q4.37 In a head-on collision between a compact 1000 kg car and a large 2500 kg car, which one experiences the greater force? Explain. Which one experiences the greater acceleration? Explain why. Why are passengers in the small car more likely to be injured than those in the large car, even when the two car bodies are equally strong?

Q4.38 Suppose you are in a rocket with no windows, traveling in deep space far from other objects. Without looking outside the rocket or making any contact with the outside world, explain how you could determine whether the rocket is (a) moving forward at a constant 80% of the speed of light and (b) accelerating in the forward direction.

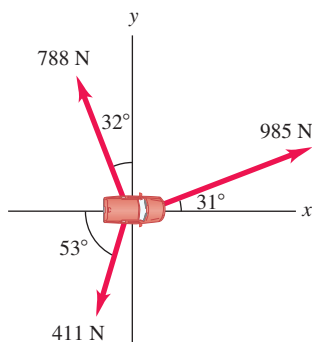
EXERCISES

Section 4.1 Force and Interactions

4.1 •• Two dogs pull horizontally on ropes attached to a post; the angle between the ropes is 60.0° . If Rover exerts a force of 270 N and Fido exerts a force of 300 N, find the magnitude of the resultant force and the angle it makes with Rover's rope.

4.2 • To extricate an SUV stuck in the mud, workmen use three horizontal ropes, producing the force vectors shown in **Fig. E4.2**. (a) Find the x - and y -components of each of the three pulls. (b) Use the components to find the magnitude and direction of the resultant of the three pulls.

Figure E4.2



4.3 • BIO Jaw Injury. Due to a jaw injury, a patient must wear a strap (**Fig. E4.3**) that produces a net upward force of 5.00 N on his chin. The tension is the same throughout the strap. To what tension must the strap be adjusted to provide the necessary upward force?

4.4 • A man is dragging a trunk up the loading ramp of a mover's truck. The ramp has a slope angle of 20.0° , and the man pulls upward with a force $\vec{F}$ whose direction makes an angle of 30.0° with the ramp (**Fig. E4.4**). (a) How large a force $\vec{F}$ is necessary for the component F_x parallel to the ramp to be 90.0 N? (b) How large will the component F_y perpendicular to the ramp be then?

Figure E4.4

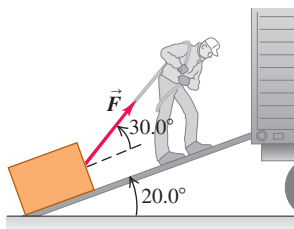
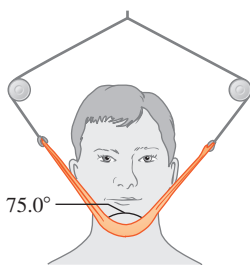


Figure E4.3



4.5 • Forces $\vec{F}_1$ and $\vec{F}_2$ act at a point. The magnitude of $\vec{F}_1$ is 9.00 N, and its direction is 60.0° above the x -axis in the second quadrant. The magnitude of $\vec{F}_2$ is 6.00 N, and its direction is 53.1° below the x -axis in the third quadrant. (a) What are the x - and y -components of the resultant force? (b) What is the magnitude of the resultant force?

Section 4.3 Newton's Second Law

4.6 • An electron (mass = 9.11×10^{-31} kg) leaves one end of a TV picture tube with zero initial speed and travels in a straight line to the accelerating grid, which is 1.80 cm away. It reaches the grid with a speed of 3.00×10^6 m/s. If the accelerating force is constant, compute (a) the acceleration; (b) the time to reach the grid; and (c) the net force, in newtons. Ignore the gravitational force on the electron.

4.7 •• A 68.5 kg skater moving initially at 2.40 m/s on rough horizontal ice comes to rest uniformly in 3.52 s due to friction from the ice. What force does friction exert on the skater?

4.8 •• You walk into an elevator, step onto a scale, and push the "up" button. You recall that your normal weight is 625 N. Draw a free-body diagram. (a) When the elevator has an upward acceleration of magnitude 2.50 m/s², what does the scale read? (b) If you hold a 3.85 kg package by a light vertical string, what will be the tension in this string when the elevator accelerates as in part (a)?

4.9 • A box rests on a frozen pond, which serves as a frictionless horizontal surface. If a fisherman applies a horizontal force with magnitude 48.0 N to the box and produces an acceleration of magnitude 2.20 m/s², what is the mass of the box?

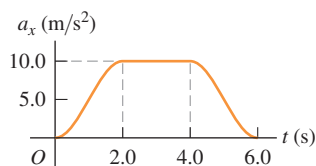
4.10 •• A dockworker applies a constant horizontal force of 80.0 N to a block of ice on a smooth horizontal floor. The frictional force is negligible. The block starts from rest and moves 11.0 m in 5.00 s. (a) What is the mass of the block of ice? (b) If the worker stops pushing at the end of 5.00 s, how far does the block move in the next 5.00 s?

4.11 • A hockey puck with mass 0.160 kg is at rest at the origin ($x = 0$) on the horizontal, frictionless surface of the rink. At time $t = 0$ a player applies a force of 0.250 N to the puck, parallel to the x -axis; she continues to apply this force until $t = 2.00$ s. (a) What are the position and speed of the puck at $t = 2.00$ s? (b) If the same force is again applied at $t = 5.00$ s, what are the position and speed of the puck at $t = 7.00$ s?

4.12 • A crate with mass 32.5 kg initially at rest on a warehouse floor is acted on by a net horizontal force of 14.0 N. (a) What acceleration is produced? (b) How far does the crate travel in 10.0 s? (c) What is its speed at the end of 10.0 s?

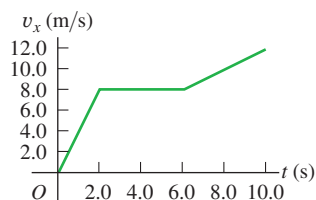
4.13 • A 4.50 kg experimental cart undergoes an acceleration in a straight line (the x -axis). The graph in **Fig. E4.13** shows this acceleration as a function of time. (a) Find the maximum net force on this cart. When does this maximum force occur? (b) During what times is the net force on the cart a constant? (c) When is the net force equal to zero?

Figure E4.13



4.14 • A 2.75 kg cat moves in a straight line (the x -axis). **Figure E4.14** shows a graph of the x -component of this cat's velocity as a function of time. (a) Find the maximum net force on this cat. When does this force occur? (b) When is the net force on the cat equal to zero? (c) What is the net force at time 8.5 s?

Figure E4.14



4.15 • A small 8.00 kg rocket burns fuel that exerts a time-varying upward force on the rocket (assume constant mass) as the rocket moves upward from the launch pad. This force obeys the equation $F = A + Bt^2$. Measurements show that at $t = 0$, the force is 100.0 N, and at the end of the first 2.00 s, it is 150.0 N. (a) Find the constants A and B , including their SI units. (b) Find the *net* force on this rocket and its acceleration (i) the instant after the fuel ignites and (ii) 3.00 s after the fuel ignites. (c) Suppose that you were using this rocket in outer space, far from all gravity. What would its acceleration be 3.00 s after fuel ignition?

Section 4.4 Mass and Weight

4.16 • An astronaut's pack weighs 17.5 N when she is on the earth but only 3.24 N when she is at the surface of a moon. (a) What is the acceleration due to gravity on this moon? (b) What is the mass of the pack on this moon?

4.17 • Superman throws a 2400 N boulder at an adversary. What horizontal force must Superman apply to the boulder to give it a horizontal acceleration of 12.0 m/s²?

4.18 • BIO (a) An ordinary flea has a mass of 210 μg . How many newtons does it weigh? (b) The mass of a typical frog hopper is 12.3 mg. How many newtons does it weigh? (c) A house cat typically weighs 45 N. How many pounds does it weigh, and what is its mass in kilograms?

4.19 • At the surface of Jupiter's moon Io, the acceleration due to gravity is $g = 1.81 \text{ m/s}^2$. A watermelon weighs 44.0 N at the surface of the earth. (a) What is the watermelon's mass on the earth's surface? (b) What would be its mass and weight on the surface of Io?

Section 4.5 Newton's Third Law

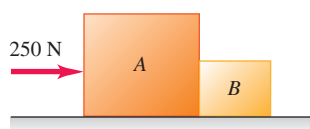
4.20 • Estimate the mass in kilograms and the weight in pounds of a typical sumo wrestler. How do your estimates for the wrestler compare to your estimates of the average mass and weight of the students in your physics class? Do a web search if necessary to help make the estimates. In your solution list what values you assume for the quantities you use in making your estimates.

4.21 • BIO World-class sprinters can accelerate out of the starting blocks with an acceleration that is nearly horizontal and has magnitude 15 m/s². How much horizontal force must a 55 kg sprinter exert on the starting blocks to produce this acceleration? Which object exerts the force that propels the sprinter: the blocks or the sprinter herself?

4.22 • A small car of mass 380 kg is pushing a large truck of mass 900 kg due east on a level road. The car exerts a horizontal force of 1600 N on the truck. What is the magnitude of the force that the truck exerts on the car?

4.23 •• Boxes A and B are in contact on a horizontal, frictionless surface (**Fig. E4.23**). Box A has mass 20.0 kg and box B has mass 5.0 kg. A horizontal force of 250 N is exerted on box A . What is the magnitude of the force that box A exerts on box B ?

Figure E4.23



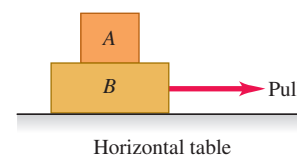
4.24 •• The upward normal force exerted by the floor is 620 N on an elevator passenger who weighs 650 N. What are the reaction forces to these two forces? Is the passenger accelerating? If so, what are the magnitude and direction of the acceleration?

4.25 •• A student of mass 45 kg jumps off a high diving board. What is the acceleration of the earth toward her as she accelerates toward the earth with an acceleration of 9.8 m/s²? Use $6.0 \times 10^{24} \text{ kg}$ for the mass of the earth, and assume that the net force on the earth is the force of gravity she exerts on it.

Section 4.6 Free-Body Diagrams

4.26 •• You pull horizontally on block B in **Fig. E4.26**, causing both blocks to move together as a unit. For this moving system, make a carefully labeled free-body diagram of block A if (a) the table is frictionless and (b) there is friction between block B and the table and the pull is equal in magnitude to the friction force on block B due to the table.

Figure E4.26



4.27 •• Crates A and B sit at rest side by side on a frictionless horizontal surface. They have masses m_A and m_B , respectively. When a horizontal force $\vec{F}$ is applied to crate A , the two crates move off to the right. (a) Draw clearly labeled free-body diagrams for crate A and for crate B . Indicate which pairs of forces, if any, are third-law action–reaction pairs. (b) If the magnitude of $\vec{F}$ is less than the total weight of the two crates, will it cause the crates to move? Explain.

4.28 •• CP A .22 caliber rifle bullet traveling at 350 m/s strikes a large tree and penetrates it to a depth of 0.130 m. The mass of the bullet is 1.80 g. Assume a constant retarding force. (a) How much time is required for the bullet to stop? (b) What force, in newtons, does the tree exert on the bullet?

4.29 • A ball is hanging from a long string that is tied to the ceiling of a train car traveling eastward on horizontal tracks. An observer inside the train car sees the ball hang motionless. Draw a clearly labeled free-body diagram for the ball if (a) the train has a uniform velocity and (b) the train is speeding up uniformly. Is the net force on the ball zero in either case? Explain.

4.30 •• A chair of mass 12.0 kg is sitting on the horizontal floor; the floor is not frictionless. You push on the chair with a force $F = 40.0 \text{ N}$ that is directed at an angle of 37.0° below the horizontal, and the chair slides along the floor. (a) Draw a clearly labeled free-body diagram for the chair. (b) Use your diagram and Newton's laws to calculate the normal force that the floor exerts on the chair.

PROBLEMS

4.31 •• CP Estimate the average force that a major-league pitcher exerts on the baseball when he throws a fastball. Express your answer in pounds. In your solution, list the quantities for which you estimate values and any assumptions you make. Do a web search to help determine the values you use in making your estimates.

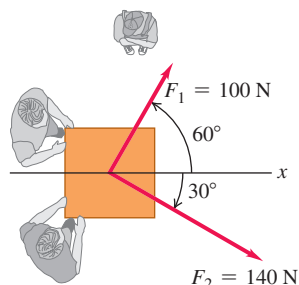
4.32 •• CP You have just landed on Planet X. You release a 100 g ball from rest from a height of 10.0 m and measure that it takes 3.40 s to reach the ground. Ignore any force on the ball from the atmosphere of the planet. How much does the 100 g ball weigh on the surface of Planet X?

4.33 •• CP A 5.60 kg bucket of water is accelerated upward by a cord of negligible mass whose breaking strength is 75.0 N. If the bucket starts from rest, what is the minimum time required to raise the bucket a vertical distance of 12.0 m without breaking the cord?

4.34 •• Block A rests on top of block B as shown in **Fig. E4.26**. The table is frictionless but there is friction (a horizontal force) between blocks A and B . Block B has mass 6.00 kg and block A has mass 2.00 kg. If the horizontal pull applied to block B equals 12.0 N, then block B has an acceleration of 1.80 m/s². What is the acceleration of block A ?

4.35 •• Two adults and a child want to push a wheeled cart in the direction marked x in **Fig. P4.35** (next page). The two adults push with horizontal forces $\vec{F}_1$ and $\vec{F}_2$ as shown. (a) Find the magnitude and direction of the *smallest* force that the child should exert. Ignore the effects of friction. (b) If the child exerts the minimum force found in part (a), the cart accelerates at 2.0 m/s² in the $+x$ -direction. What is the weight of the cart?

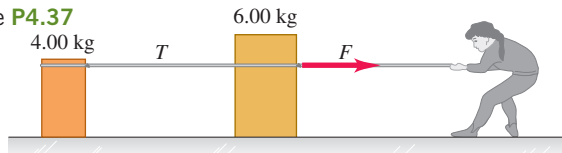
Figure P4.35



4.36 ••• CP An advertisement claims that a particular automobile can “stop on a dime.” What net force would be necessary to stop a 850 kg automobile traveling initially at 45.0 km/h in a distance equal to the diameter of a dime, 1.8 cm?

4.37 • Two crates, one with mass 4.00 kg and the other with mass 6.00 kg, sit on the frictionless surface of a frozen pond, connected by a light rope (**Fig. P4.37**). A woman wearing golf shoes (for traction) pulls horizontally on the 6.00 kg crate with a force F that gives the crate an acceleration of 2.50 m/s^2 . (a) What is the acceleration of the 4.00 kg crate? (b) Draw a free-body diagram for the 4.00 kg crate. Use that diagram and Newton’s second law to find the tension T in the rope that connects the two crates. (c) Draw a free-body diagram for the 6.00 kg crate. What is the direction of the net force on the 6.00 kg crate? Which is larger in magnitude, T or F ? (d) Use part (c) and Newton’s second law to calculate the magnitude of F .

Figure P4.37



4.38 •• CP Two blocks connected by a light horizontal rope sit at rest on a horizontal, frictionless surface. Block A has mass 15.0 kg, and block B has mass m . A constant horizontal force $F = 60.0 \text{ N}$ is applied to block A (**Fig. P4.38**). In the first 5.00 s after the force is applied, block A moves 18.0 m to the right. (a) While the blocks are moving, what is the tension T in the rope that connects the two blocks? (b) What is the mass of block B?

Figure P4.38



4.39 • CALC To study damage to aircraft that collide with large birds, you design a test gun that will accelerate chicken-sized objects so that their displacement along the gun barrel is given by $x = (9.0 \times 10^3 \text{ m/s}^2)t^2 - (8.0 \times 10^4 \text{ m/s}^3)t^3$. The object leaves the end of the barrel at $t = 0.025 \text{ s}$. (a) How long must the gun barrel be? (b) What will be the speed of the objects as they leave the end of the barrel? (c) What net force must be exerted on a 1.50 kg object at (i) $t = 0$ and (ii) $t = 0.025 \text{ s}$?

4.40 •• CP On a test flight a rocket with mass 400 kg blasts off from the surface of the earth. The rocket engines apply a constant upward force F until the rocket reaches a height of 100 m and then they shut off. If the rocket is to reach a maximum height of 400 m above the surface of the earth, what value of F is required? Assume the change in the rocket’s mass is negligible.

4.41 •• CP After an annual checkup, you leave your physician’s office, where you weighed 683 N. You then get into an elevator that, conveniently, has a scale. Find the magnitude and direction of the elevator’s acceleration if the scale reads (a) 725 N and (b) 595 N.

4.42 • A loaded elevator with very worn cables has a total mass of 2200 kg, and the cables can withstand a maximum tension of 28,000 N. (a) Draw the free-body force diagram for the elevator. In terms of the forces on your diagram, what is the net force on the elevator? Apply Newton’s second law to the elevator and find the maximum upward acceleration for the elevator if the cables are not to break. (b) What would be the answer to part (a) if the elevator were on the moon, where $g = 1.62 \text{ m/s}^2$?

4.43 •• CP A batter swings at a baseball (mass 0.145 kg) that is moving horizontally toward him at a speed of 40.0 m/s. He hits a line drive with the ball moving away from him horizontally at 50.0 m/s just after it leaves the bat. If the bat and ball are in contact for 8.00 ms, what is the average force that the bat applies to the ball?

4.44 •• CP CALC An object with mass m is moving along the x -axis according to the equation $x(t) = \alpha t^2 - 2\beta t$, where α and β are positive constants. What is the magnitude of the net force on the object at time $t = 0$?

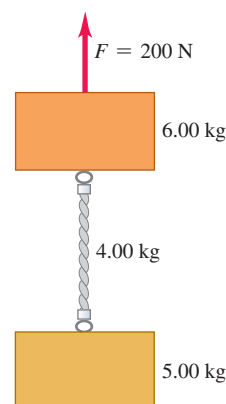
4.45 •• CP Boxes A and B are connected to each end of a light vertical rope (**Fig. P4.45**). A constant upward force $F = 80.0 \text{ N}$ is applied to box A. Starting from rest, box B descends 12.0 m in 4.00 s. The tension in the rope connecting the two boxes is 36.0 N. What are the masses of (a) box B, (b) box A?

Figure P4.45



4.46 •• The two blocks in **Fig. P4.46** are connected by a heavy uniform rope with a mass of 4.00 kg. An upward force of 200 N is applied as shown. (a) Draw three free-body diagrams: one for the 6.00 kg block, one for the 4.00 kg rope, and another one for the 5.00 kg block. For each force, indicate what object exerts that force. (b) What is the acceleration of the system? (c) What is the tension at the top of the heavy rope? (d) What is the tension at the midpoint of the rope?

Figure P4.46



4.47 ••• CP A small rocket with mass 20.0 kg is moving in free fall toward the earth. Air resistance can be neglected. When the rocket is 80.0 m above the surface of the earth, it is moving downward with a speed of 30.0 m/s. At that instant the rocket engines start to fire and produce a constant upward force F on the rocket. Assume the change in the rocket’s mass is negligible. What is the value of F if the rocket’s speed becomes zero just as it reaches the surface of the earth, for a soft landing? (*Hint:* The net force on the rocket is the combination of the upward force F from the engines and the downward weight of the rocket.)

4.48 •• CP Extraterrestrial Physics. You have landed on an unknown planet, Newtonia, and want to know what objects weigh there. When you push a certain tool, starting from rest, on a frictionless horizontal surface with a 12.0 N force, the tool moves 16.0 m in the first 2.00 s. You next observe that if you release this tool from rest at 10.0 m above the ground, it takes 2.58 s to reach the ground. What does the tool weigh on Newtonia, and what does it weigh on earth?

4.49 •• CP CALC A mysterious rocket-propelled object of mass 45.0 kg is initially at rest in the middle of the horizontal, frictionless surface of an ice-covered lake. Then a force directed east and with magnitude $F(t) = (16.8 \text{ N/s})t$ is applied. How far does the object travel in the first 5.00 s after the force is applied?

4.50 •• CP Starting at time $t = 0$, net force F_1 is applied to an object that is initially at rest. (a) If the force remains constant with magnitude F_1 while the object moves a distance d , the final speed of the object is v_1 . What is the final speed v_2 (in terms of v_1) if the net force is $F_2 = 2F_1$ and the object moves the same distance d while the force is being applied? (b) If the force F_1 remains constant while it is applied for a time T , the final speed of the object is v_1 . What is the final speed v_2 (in terms of v_1) if the applied force is $F_2 = 2F_1$ and is constant while it is applied for the same time T ? In a later chapter we'll call force times distance work and force times time impulse and associate work and impulse with the change in speed.)

4.51 •• DATA The table* gives automobile performance data for a few types of cars:

Make and Model (Year)	Mass (kg)	Time (s) to go from 0 to 60 mph
Alpha Romeo 4C (2013)	895	4.4
Honda Civic 2.0i (2011)	1320	6.4
Ferrari F430 (2004)	1435	3.9
Ford Focus RS500 (2010)	1468	5.4
Volvo S60 (2013)	1650	7.2

*Source: www.autosnout.com

(a) During an acceleration of 0 to 60 mph, which car has the largest average net force acting on it? The smallest? (b) During this acceleration, for which car would the average net force on a 72.0 kg passenger be the largest? The smallest? (c) When the Ferrari F430 accelerates from 0 to 100 mph in 8.6 s, what is the average net force acting on it? How does this net force compare with the average net force during the acceleration from 0 to 60 mph? Explain why these average net forces might differ. (d) Discuss why a car has a top speed. What is the net force on the Ferrari F430 when it is traveling at its top speed, 196 mph?

4.52 ••• CALC The position of a training helicopter (weight $2.75 \times 10^5 \text{ N}$) in a test is given by $\vec{r} = (0.020 \text{ m/s}^3)t^3\hat{i} + (2.2 \text{ m/s})t\hat{j} - (0.060 \text{ m/s}^2)t^2\hat{k}$. Find the net force on the helicopter at $t = 5.0 \text{ s}$.

4.53 •• DATA You are a Starfleet captain going boldly where no one has gone before. You land on a distant planet and visit an engineering testing lab. In one experiment a short, light rope is attached to the top of a block and a constant upward force F is applied to the free end of the rope. The block has mass m and is initially at rest. As F is varied, the time for the block to move upward 8.00 m is measured. The values that you collected are given in the table:

F (N)	Time (s)
250	3.3
300	2.2
350	1.7
400	1.5
450	1.3
500	1.2

(a) Plot F versus the acceleration a of the block. (b) Use your graph to determine the mass m of the block and the acceleration of gravity g at the surface of the planet. Note that even on that planet, measured values contain some experimental error.

4.54 •• DATA An 8.00 kg box sits on a level floor. You give the box a sharp push and find that it travels 8.22 m in 2.8 s before coming to rest again. (a) You measure that with a different push the box traveled 4.20 m in 2.0 s. Do you think the box has a constant acceleration as it slows down? Explain your reasoning. (b) You add books to the box to increase its mass. Repeating the experiment, you give the box a push and measure how long it takes the box to come to rest and how far the box travels. The results, including the initial experiment with no added mass, are given in the table:

Added Mass (kg)	Distance (m)	Time (s)
0	8.22	2.8
3.00	10.75	3.2
7.00	9.45	3.0
12.0	7.10	2.6

In each case, did your push give the box the same initial speed? What is the ratio between the greatest initial speed and the smallest initial speed for these four cases? (c) Is the average horizontal force f exerted on the box by the floor the same in each case? Graph the magnitude of force f versus the total mass m of the box plus its contents, and use your graph to determine an equation for f as a function of m .

CHALLENGE PROBLEMS

4.55 ••• CP CALC A block of mass 2.00 kg is initially at rest at $x = 0$ on a slippery horizontal surface for which there is no friction. Starting at time $t = 0$, a horizontal force $F_x(t) = \beta - \alpha t$ is applied to the block, where $\alpha = 6.00 \text{ N/s}$ and $\beta = 4.00 \text{ N}$. (a) What is the largest positive value of x reached by the block? How long does it take the block to reach this point, starting from $t = 0$, and what is the magnitude of the force when the block is at this value of x ? (b) How long from $t = 0$ does it take the block to return to $x = 0$, and what is its speed at this point?

4.56 ••• CALC An object of mass m is at rest in equilibrium at the origin. At $t = 0$ a new force $\vec{F}(t)$ is applied that has components

$$F_x(t) = k_1 + k_2y \quad F_y(t) = k_3t$$

where k_1 , k_2 , and k_3 are constants. Calculate the position $\vec{r}(t)$ and velocity $\vec{v}(t)$ vectors as functions of time.

MCAT-STYLE PASSAGE PROBLEMS

BIO Forces on a Dancer's Body. Dancers experience large forces associated with the jumps they make. For example, when a dancer lands after a vertical jump, the force exerted on the head by the neck must exceed the head's weight by enough to cause the head to slow down and come to rest. The head is about 9.4% of a typical person's mass. Video analysis of a 65 kg dancer landing after a vertical jump shows that her head decelerates from 4.0 m/s to rest in a time of 0.20 s.

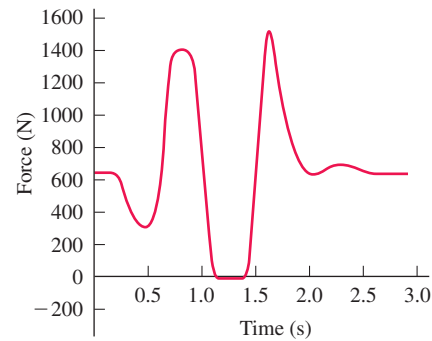
4.57 What is the magnitude of the average force that her neck exerts on her head during the landing? (a) 0 N; (b) 60 N; (c) 120 N; (d) 180 N.

4.58 Compared with the force her neck exerts on her head during the landing, the force her head exerts on her neck is (a) the same; (b) greater; (c) smaller; (d) greater during the first half of the landing and smaller during the second half of the landing.

4.59 While the dancer is in the air and holding a fixed pose, what is the magnitude of the force her neck exerts on her head? (a) 0 N; (b) 60 N; (c) 120 N; (d) 180 N.

4.60 The forces on a dancer can be measured directly when a dancer performs a jump on a force plate that measures the force between her feet and the ground. A graph of force versus time throughout a vertical jump performed on a force plate is shown in **Fig. P4.60**. What is happening at 0.4 s? The dancer is (a) bending her legs so that her body is accelerating downward; (b) pushing her body up with her legs and is almost ready to leave the ground; (c) in the air and at the top of her jump; (d) landing and her feet have just touched the ground.

Figure P4.60



ANSWERS

Chapter Opening Question ?

(v) Newton's third law tells us that the barbell pushes on the weightlifter just as hard as the weightlifter pushes on the barbell in *all* circumstances, no matter how the barbell is moving. However, the magnitude of the force that the weightlifter exerts is different in different circumstances. This force magnitude is equal to the weight of the barbell when the barbell is stationary, moving upward at a constant speed, or moving downward at a constant speed; it is greater than the weight of the barbell when the barbell accelerates upward; and it is less than the weight of the barbell when the barbell accelerates downward. But in each case the push of the barbell on the weightlifter has exactly the same magnitude as the push of the weightlifter on the barbell.

Key Example **√**ARIATION Problems

VP4.1.1 $\Sigma F_x = 88.0 \text{ N}$, $\Sigma F_y = -44.0 \text{ N}$, $F = 98.4 \text{ N}$, angle = 26.6° clockwise from the $+x$ -direction

VP4.1.2 $\Sigma F_x = -143 \text{ N}$, $\Sigma F_y = 246 \text{ N}$, $F = 285 \text{ N}$, angle = 120° counterclockwise from the $+x$ -direction

VP4.1.3 $\Sigma F_x = 6.4 \text{ N}$, $\Sigma F_y = 1.5 \text{ N}$, $F = 6.6 \text{ N}$, angle = 13° counterclockwise from the $+x$ -direction

VP4.1.4 $F_{\text{Tsureku}, x} = -55.8 \text{ N}$, $F_{\text{Tsureku}, y} = -62.0 \text{ N}$, magnitude $F_{\text{Tsureku}} = 83.4 \text{ N}$, angle = 48.0° south of west

VP4.4.1 0.45 m/s^2

VP4.4.2 (a) normal force, exerted by the floor; weight or gravitational force, exerted by the earth; horizontal force, exerted by your hand
(b) 0.25 m/s^2

VP4.4.3 (a) 12 N (b) equal

VP4.4.4 (a) 9.60 N (b) 4.4 N, in the direction opposite to your push

VP4.5.1 40.0 N

VP4.5.2 $1.65 \times 10^3 \text{ m}$

VP4.5.3 (a) 3.84 s (b) 6.51 m/s^2 (c) $1.56 \times 10^4 \text{ N}$

VP4.5.4 (a) gravity (the earth), normal force (the road), friction (the road), push (you and your friends) (b) 575 N

Bridging Problem

(a) See the Video Tutor Solution in Mastering Physics.

(b) (i) 2.20 m/s^2 ; (ii) 6.00 N; (iii) 3.00 N